

Once again, an owner wishes to hire a manager to run a one-time project. Now, however, the manager's effort level, denoted by  $e$ , is fully observable. What is not observable after the contract is signed is the random realization of the manager's disutility from effort. For example, the manager may come to find himself well suited to the tasks required at the firm, in which case high effort has a relatively low disutility associated with it, or the opposite may be true. However, only the manager comes to know which case obtains.<sup>10</sup>

Before proceeding, we note that the techniques we develop here can also be applied to models of *monopolistic screening* where, in a setting characterized by *precontractual* informational asymmetries, a single uninformed individual offers a menu of contracts in order to distinguish, or *screen*, informed agents who have differing information at the time of contracting (see Section 13.D for an analysis of a competitive screening model). We discuss this connection further in small type at the end of this section.

To formulate our hidden information principal-agent model, we suppose that effort can be measured by a one-dimensional variable  $e \in [0, \infty)$ . Gross profits (excluding any wage payments to the manager) are a simple deterministic function of effort,  $\pi(e)$ , with  $\pi(0) = 0$ ,  $\pi'(e) > 0$ , and  $\pi''(e) < 0$  for all  $e$ .

The manager is an expected utility maximizer whose Bernoulli utility function over wages and effort,  $u(w, e, \theta)$ , depends on a state of nature  $\theta$  that is realized after the contract is signed and that only the manager observes. We assume that  $\theta \in \mathbb{R}$ , and we focus on a special form of  $u(w, e, \theta)$  that is widely used in the literature:<sup>11</sup>

$$u(w, e, \theta) = v(w - g(e, \theta)).$$

The function  $g(e, \theta)$  measures the disutility of effort in monetary units. We assume that  $g(0, \theta) = 0$  for all  $\theta$  and, letting subscripts denote partial derivatives, that

$$g_e(e, \theta) \begin{cases} > 0 & \text{for } e > 0 \\ = 0 & \text{for } e = 0 \end{cases}$$

$$g_{ee}(e, \theta) < 0 \quad \text{for all } e$$

$$g_\theta(e, \theta) < 0 \quad \text{for all } e$$

$$g_{e\theta}(e, \theta) \begin{cases} < 0 & \text{for } e > 0 \\ = 0 & \text{for } e = 0. \end{cases}$$

Thus, the manager is averse to increases in effort, and this aversion is larger the greater the current level of effort. In addition, higher values of  $\theta$  are more productive states in the sense that both the manager's total disutility from effort,  $g(e, \theta)$ , and his marginal disutility from effort at any current effort level,  $g_e(e, \theta)$ , are lower when  $\theta$

10. A seemingly more important source of hidden information between managers and owners is that the manager of a firm often comes to know more about the potential profitability of various actions than does the owner. In Section 14.D, we discuss one hybrid hidden action-hidden information model that captures this alternative sort of informational asymmetry; its formal analysis reduces to that of the model studied here.

11. Exercise 14.C.3 asks you to consider an alternative form for the manager's utility function.

is greater. We also assume that the manager is strictly risk averse, with  $v''(\cdot) < 0$ .<sup>12</sup> As in Section 14.B, the manager's reservation utility level, the level of expected utility he must receive if he is to accept the owner's contract offer, is denoted by  $\bar{u}$ . Note that our assumptions about  $g(e, \theta)$  imply that the manager's indifference curves have the single-crossing property discussed in Section 13.C.

Finally, for expositional purposes, we focus on the simple case in which  $\theta$  can take only one of two values,  $\theta_H$  and  $\theta_L$ , with  $\theta_H > \theta_L$  and  $\text{Prob}(\theta_H) = \lambda \in (0, 1)$ . (Exercise 14.C.1 asks you to consider the case of an arbitrary finite number of states.)

A contract must try to accomplish two objectives here: first, as in Section 14.B, the risk-neutral owner should insure the manager against fluctuations in his income; second, although there is no problem here in insuring that the manager puts in effort (because the contract can explicitly state the effort level required), a contract that maximizes the surplus available in the relationship (and hence, the owner's payoff) must make the level of managerial effort responsive to the disutility incurred by the manager, that is, to the state  $\theta$ . To fix ideas, we first illustrate how these goals are accomplished when  $\theta$  is observable; we then turn to an analysis of the problems that arise when  $\theta$  is observed only by the manager.

### *The State $\theta$ is Observable*

If  $\theta$  is observable, a contract can directly specify the effort level and remuneration of the manager contingent on each realization of  $\theta$  (note that these variables fully determine the economic outcomes for the two parties). Thus, a complete information contract consists of two wage-effort pairs:  $(w_H, e_H) \in \mathbb{R} \times \mathbb{R}_+$  for state  $\theta_H$  and  $(w_L, e_L) \in \mathbb{R} \times \mathbb{R}_+$  for state  $\theta_L$ . The owner optimally chooses these pairs to solve the following problem:

$$\begin{aligned} \text{Max}_{\substack{w_L, e_L \geq 0 \\ w_H, e_H \geq 0}} \quad & \lambda[\pi(e_H) - w_H] + (1 - \lambda)[\pi(e_L) - w_L] \end{aligned} \quad (14.C.1)$$

$$\text{s.t.} \quad \lambda v(w_H - g(e_H, \theta_H)) + (1 - \lambda)v(w_L - g(e_L, \theta_L)) \geq \bar{u}.$$

In any solution  $[(w_L^*, e_L^*), (w_H^*, e_H^*)]$  to problem (14.C.1) the reservation utility constraint must bind; otherwise, the owner could lower the level of wages offered and still have the manager accept the contract. In addition, letting  $\gamma \geq 0$  denote the multiplier on this constraint, the solution must satisfy the following first-order conditions:

$$-\lambda + \gamma \lambda v'(w_H^* - g(e_H^*, \theta_H)) = 0. \quad (14.C.2)$$

$$-(1 - \lambda) + \gamma(1 - \lambda) v'(w_L^* - g(e_L^*, \theta_L)) = 0. \quad (14.C.3)$$

$$\lambda \pi'(e_H^*) - \gamma \lambda v'(w_H^* - g(e_H^*, \theta_H)) g_e(e_H^*, \theta_H) \begin{cases} \leq 0, \\ = 0 \end{cases} \quad \text{if } e_H^* > 0. \quad (14.C.4)$$

$$(1 - \lambda) \pi'(e_L^*) - \gamma(1 - \lambda) v'(w_L^* - g(e_L^*, \theta_L)) g_e(e_L^*, \theta_L) \begin{cases} \leq 0, \\ = 0 \end{cases} \quad \text{if } e_L^* > 0. \quad (14.C.5)$$

12. As with the case of hidden actions studied in Section 14.B, nonobservability causes no welfare loss in the case of managerial risk neutrality. As there, a "sellout" contract that faces the manager with the full marginal returns from his actions can generate the first-best outcome. (See Exercise 14.C.2.)

These conditions indicate how the two objectives of insuring the manager and making effort sensitive to the state are handled. First, rearranging and combining conditions (14.C.2) and (14.C.3), we see that

$$v'(w_H^* - g(e_H^*, \theta_H)) = v'(w_L^* - g(e_L^*, \theta_L)), \quad (14.C.6)$$

so the manager's marginal utility of income is equalized across states. This is the usual condition for a risk-neutral party optimally insuring a risk-averse individual. Condition (14.C.6) implies that  $w_H^* - g(e_H^*, \theta_H) = w_L^* - g(e_L^*, \theta_L)$ , which in turn implies that  $v(w_H^* - g(e_H^*, \theta_H)) = v(w_L^* - g(e_L^*, \theta_L))$ ; that is, the manager's utility is equalized across states. Given the reservation utility constraint in (14.C.1), the manager therefore has utility level  $\bar{u}$  in each state.

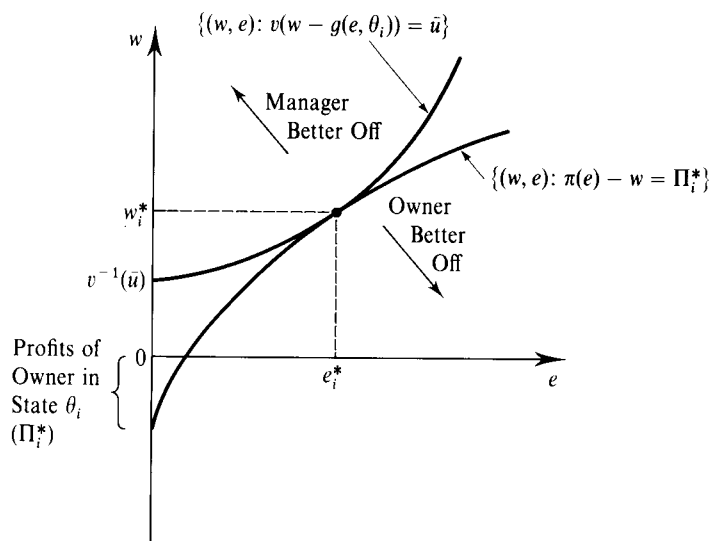
Now consider the optimal effort levels in the two states. Since  $g_e(0, \theta) = 0$  and  $\pi'(0) > 0$ , conditions (14.C.4) and (14.C.5) must hold with equality and  $e_i^* > 0$  for  $i = 1, 2$ . Combining condition (14.C.2) with (14.C.4), and condition (14.C.3) with (14.C.5), we see that the optimal level of effort in state  $\theta_i$ ,  $e_i^*$ , satisfies

$$\pi'(e_i^*) = g_e(e_i^*, \theta_i) \quad \text{for } i = L, H. \quad (14.C.7)$$

This condition says that the optimal level of effort in state  $\theta_i$  equates the marginal benefit of effort in terms of increased profit with its marginal disutility cost.

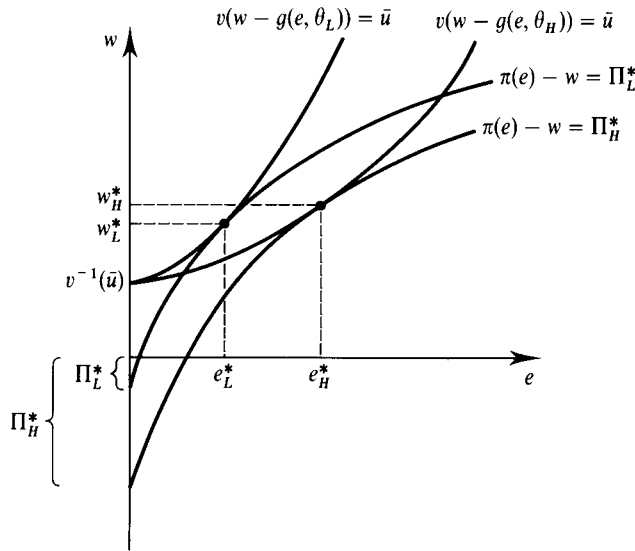
The pair  $(w_i^*, e_i^*)$  is illustrated in Figure 14.C.1 (note that the wage is depicted on the vertical axis and the effort level on the horizontal axis). As shown, the manager is better off as we move to the northwest (higher wages and less effort), and the owner is better off as we move toward the southeast. Because the manager receives utility level  $\bar{u}$  in state  $\theta_i$ , the owner seeks to find the most profitable point on the manager's state  $\theta_i$  indifference curve with utility level  $\bar{u}$ . This is a point of tangency between the manager's indifference curve and one of the owner's isoprofit curves. At this point, the marginal benefit to additional effort in terms of increased profit is exactly equal to the marginal cost borne by the manager.

The owner's profit level in state  $\theta_i$  is  $\Pi_i^* = \pi(e_i^*) - v^{-1}(\bar{u}) - g(e_i^*, \theta_i)$ . As shown in Figure 14.C.1, this profit is exactly equal to the distance from the origin to the point at which the owner's isoprofit curve through point  $(w_i^*, e_i^*)$  hits the vertical



**Figure 14.C.1**

The optimal wage-effort pair for state  $\theta_i$  when states are observable.

**Figure 14.C.2**

The optimal contract with full observability of  $\theta$ .

axis [since  $\pi(0) = 0$ , if the wage payment at this point on the vertical axis is  $\hat{w} < 0$ , the owner's profit at  $(w_i^*, e_i^*)$  is exactly  $-\hat{w}$ ].

From condition (14.C.7), we see that  $g_{ee}(e, \theta) < 0$ ,  $\pi''(e) < 0$ , and  $g_{ee}(e, \theta) > 0$  imply that  $e_H^* > e_L^*$ . Figure 14.C.2 depicts the optimal contract,  $[(w_H^*, e_H^*), (w_L^*, e_L^*)]$ .

These observations are summarized in Proposition 14.C.1.

**Proposition 14.C.1:** In the principal-agent model with an observable state variable  $\theta$ , the optimal contract involves an effort level  $e_i^*$  in state  $\theta_i$  such that  $\pi'(e_i^*) = g_e(e_i^*, \theta_i)$  and fully insures the manager, setting his wage in each state  $\theta_i$  at the level  $w_i^*$  such that  $v(w_i^* - g(e_i^*, \theta_i)) = \bar{u}$ .

Thus, with a strictly risk-averse manager, the first-best contract is characterized by two basic features: first, the owner fully insures the manager against risk; second, he requires the manager to work to the point at which the marginal benefit of effort exactly equals its marginal cost. Because the marginal cost of effort is lower in state  $\theta_H$  than in state  $\theta_L$ , the contract calls for more effort in state  $\theta_H$ .

### *The State $\theta$ is Observed Only by the Manager*

As in Section 14.B, the desire both to insure the risk-averse manager and to elicit the proper levels of effort come into conflict when informational asymmetries are present. Suppose, for example, that the owner offers a risk-averse manager the contract depicted in Figure 14.C.2 and relies on the manager to reveal the state voluntarily. If so, the owner will run into problems. As is evident in the figure, in state  $\theta_H$ , the manager prefers point  $(w_L^*, e_L^*)$  to point  $(w_H^*, e_H^*)$ . Consequently, in state  $\theta_H$  he will lie to the owner, claiming that it is actually state  $\theta_L$ . As is also evident in the figure, this misrepresentation lowers the owner's profit.

Given this problem, what is the optimal contract for the owner to offer? To answer this question, it is necessary to start by identifying the set of possible contracts that the owner can offer. One can imagine many different forms that a contract could conceivably take. For example, the owner might offer a compensation function  $w(\pi)$  that pays the manager as a function of realized profit and that leaves the effort

choice in each state to the manager's discretion. Alternatively, the owner could offer a compensation schedule  $w(\pi)$  but restrict the possible effort choices by the manager to some degree. Another possibility is that the owner could offer compensation as a function of the observable effort level chosen by the manager, possibly again with some restriction on the allowable choices. Finally, more complicated arrangements might be imagined. For example, the manager might be required to make an announcement about what the state is and then be free to choose his effort level while facing a compensation function  $w(\pi|\hat{\theta})$  that depends on his announcement  $\hat{\theta}$ .

Although finding an optimal contract from among all these possibilities may seem a daunting task, an important result known as the *revelation principle* greatly simplifies the analysis of these types of contracting problems:<sup>13</sup>

**Proposition 14.C.2: (The Revelation Principle)** Denote the set of possible states by  $\Theta$ . In searching for an optimal contract, the owner can without loss restrict himself to contracts of the following form:

- (i) After the state  $\theta$  is realized, the manager is required to announce which state has occurred.
- (ii) The contract specifies an outcome  $[w(\hat{\theta}), e(\hat{\theta})]$  for each possible announcement  $\hat{\theta} \in \Theta$ .
- (iii) In every state  $\theta \in \Theta$ , the manager finds it optimal to report the state *truthfully*.

A contract that asks the manager to announce the state  $\theta$  and associates outcomes with the various possible announcements is known as a *revelation mechanism*. The revelation principle tells us that the owner can restrict himself to using a revelation mechanism for which the manager always responds truthfully; revelation mechanisms with this truthfulness property are known as *incentive compatible* (or *truthful*) revelation mechanisms. The revelation principle holds in an extremely wide array of incentive problems. Although we defer its formal (and very general) proof to Chapter 23 (see Sections 23.C and 23.D), its basic idea is relatively straightforward.

For example, imagine that the owner is offering a contract with a compensation schedule  $w(\pi)$  that leaves the choice of effort up to the manager. Let the resulting levels of effort in states  $\theta_L$  and  $\theta_H$  be  $e_L$  and  $e_H$ , respectively. We can now show that there is a truthful revelation mechanism that generates exactly the same outcome as this contract. In particular, suppose that the owner uses a revelation mechanism that assigns outcome  $[w(\pi(e_L)), e_L]$  if the manager announces that the state is  $\theta_L$  and outcome  $[w(\pi(e_H)), e_H]$  if the manager announces that the state is  $\theta_H$ . Consider the manager's incentives for truth telling when facing this revelation mechanism. Suppose, first, that the state is  $\theta_L$ . Under the initial contract with compensation schedule  $w(\pi)$ , the manager could have achieved outcome  $[w(\pi(e_H)), e_H]$  in state  $\theta_L$  by choosing effort level  $e_H$ . Since he instead chose  $e_L$ , it must be that in state  $\theta_L$  outcome  $[w(\pi(e_L)), e_L]$  is at least as good for the manager as outcome  $[w(\pi(e_H)), e_H]$ . Thus, under the proposed revelation mechanism, the manager will find telling the truth to be an optimal response when the state is  $\theta_L$ . A similar argument applies for state  $\theta_H$ . We see therefore that this revelation mechanism results in truthful announcements

13. Two early discussions of the revelation principle are Myerson (1979) and Dasgupta, Hammond, and Maskin (1979).

by the manager and yields exactly the same outcome as the initial contract. In fact, a similar argument can be constructed for *any* initial contract (see Chapter 23), and so the owner can restrict his attention without loss to truthful revelation mechanisms.<sup>14</sup>

To simplify the characterization of the optimal contract, we restrict attention from this point on to a specific and extreme case of managerial risk aversion: *infinite* risk aversion. In particular, we take the expected utility of the manager to equal the manager's lowest utility level across the two states. Thus, for the manager to accept the owner's contract, it must be that the manager receives a utility of at least  $\bar{u}$  in each state.<sup>15</sup> As above, efficient risk sharing requires that an infinitely risk-averse manager have a utility level equal to  $\bar{u}$  in each state. If, for example, his utility is  $\bar{u}$  in one state and  $u' > \bar{u}$  in the other, then the owner's expected wage payment is larger than necessary for giving the manager an expected utility of  $\bar{u}$ .

Given this assumption about managerial risk preferences, the revelation principle allows us to write the owner's problem as follows:

$$\begin{aligned}
 & \text{Max}_{w_H, e_H \geq 0, w_L, e_L \geq 0} \quad \lambda[\pi(e_H) - w_H] + (1 - \lambda)[\pi(e_L) - w_L] & (14.C.8) \\
 & \text{s.t.} \quad \left. \begin{aligned} & \text{(i) } w_L - g(e_L, \theta_L) \geq v^{-1}(\bar{u}) \\ & \text{(ii) } w_H - g(e_H, \theta_H) \geq v^{-1}(\bar{u}) \end{aligned} \right\} \begin{array}{l} \text{reservation utility} \\ \text{(or individual rationality)} \\ \text{constraint} \end{array} \\
 & \quad \left. \begin{aligned} & \text{(iii) } w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \\ & \text{(iv) } w_L - g(e_L, \theta_L) \geq w_H - g(e_H, \theta_L) \end{aligned} \right\} \begin{array}{l} \text{incentive compatibility} \\ \text{(or truth-telling} \\ \text{or self-selection)} \\ \text{constraints.} \end{array}
 \end{aligned}$$

The pairs  $(w_H, e_H)$  and  $(w_L, e_L)$  that the contract specifies are now the wage and effort levels that result from different *announcements* of the state by the manager; that is, the outcome if the manager announces that the state is  $\theta_i$  is  $(w_i, e_i)$ . Constraints (i) and (ii) make up the *reservation utility* (or *individual rationality*) *constraint* for the infinitely risk-averse manager; if he is to accept the contract, he must be guaranteed a utility of at least  $\bar{u}$  in each state. Hence, we must have  $v(w_i - g(e_i, \theta_i)) \geq \bar{u}$  for  $i = L, H$  or, equivalently,  $w_i - g(e_i, \theta_i) \geq v^{-1}(\bar{u})$  for  $i = L, H$ . Constraints (iii) and (iv) are the *incentive compatibility* (or *truth-telling* or *self-selection*) *constraints* for the manager in states  $\theta_H$  and  $\theta_L$ , respectively. Consider, for example, constraint (iii). The

14. One restriction that we have imposed here for expositional purposes is to limit the outcomes specified following the manager's announcement to being nonstochastic (in fact, much of the literature does so as well). Randomization can sometimes be desirable in these settings because it can aid in satisfying the incentive compatibility constraints that we introduce in problem (14.C.8). See Maskin and Riley (1984a) for an example.

15. This can be thought of as the limiting case in which, starting from the concave utility function  $v(x)$ , we take the concave transformation  $v_\rho(v) = -v(x)^\rho$  for  $\rho < 0$  as the manager's Bernoulli utility function and let  $\rho \rightarrow -\infty$ . To see this, note that the manager's expected utility over the random outcome giving  $(w_H - g(e_H, \theta_H))$  with probability  $\lambda$  and  $(w_L - g(e_L, \theta_L))$  with probability  $(1 - \lambda)$  is then  $EU = -[\lambda v_H^\rho + (1 - \lambda)v_L^\rho]$ , where  $v_i = v(w_i - g(e_i, \theta_i))$  for  $i = L, H$ . This expected utility is correctly ordered by  $(-EU)^{1/\rho} = [\lambda v_H^\rho + (1 - \lambda)v_L^\rho]^{1/\rho}$ . Now as  $\rho \rightarrow -\infty$ ,  $[\lambda v_H^\rho + (1 - \lambda)v_L^\rho]^{1/\rho} \rightarrow \text{Min}\{v_H, v_L\}$  (see Exercise 3.C.6). Hence, a contract gives the manager an expected utility greater than his (certain) reservation utility if and only if  $\text{Min}\{v(w_H - g(e_H, \theta_H)), v(w_L - g(e_L, \theta_L))\} \geq \bar{u}$ .

manager's utility in state  $\theta_H$  is  $v(w_H - g(e_H, \theta_H))$  if he tells the truth, but it is  $v(w_L - g(e_L, \theta_H))$  if he instead claims that it is state  $\theta_L$ . Thus, he will tell the truth if  $w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H)$ . Constraint (iv) follows similarly.

Note that the first-best (full observability) contract depicted in Figure 14.C.2 does not satisfy the constraints of problem (14.C.8) because it violates constraint (iii).

We analyze problem (14.C.8) through a sequence of lemmas. Our arguments for these results make extensive use of graphical analysis to build intuition. An analysis of this problem using Kuhn–Tucker conditions is presented in Appendix B.

**Lemma 14.C.1:** We can ignore constraint (ii). That is, a contract is a solution to problem (14.C.8) if and only if it is the solution to the problem derived from (14.C.8) by dropping constraint (ii).

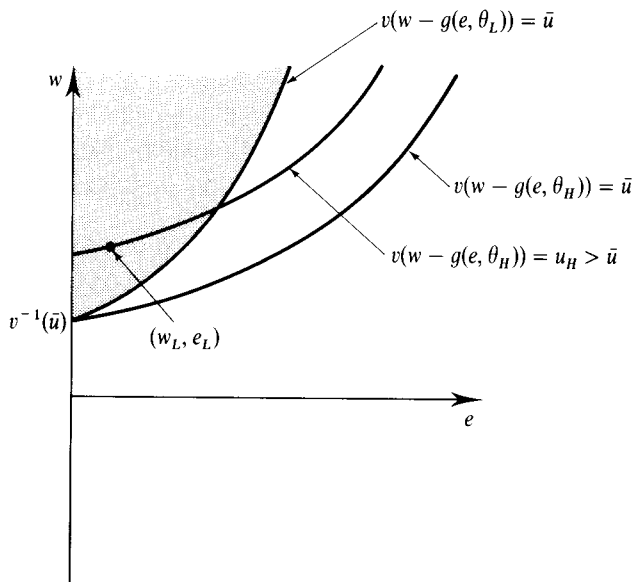
**Proof:** Whenever both constraints (i) and (iii) are satisfied, it must be that  $w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \geq w_L - g(e_L, \theta_L) \geq v^{-1}(\bar{u})$ , and so constraint (ii) is also satisfied. This implies that the set of feasible contracts in the problem derived from (14.C.8) by dropping constraint (ii) is exactly the same as the set of feasible contracts in problem (14.C.8). ■

Lemma 14.C.1 is illustrated in Figure 14.C.3. By constraint (i),  $(w_L, e_L)$  must lie in the shaded region of the figure. But by constraint (iii),  $(w_H, e_H)$  must lie on or above the state  $\theta_H$  indifference curve through point  $(w_L, e_L)$ . As can be seen, this implies that the manager's state  $\theta_H$  utility is at least  $\bar{u}$ , the utility he gets at point  $(w, e) = (v^{-1}(\bar{u}), 0)$ .

Therefore, from this point on we can ignore constraint (ii).

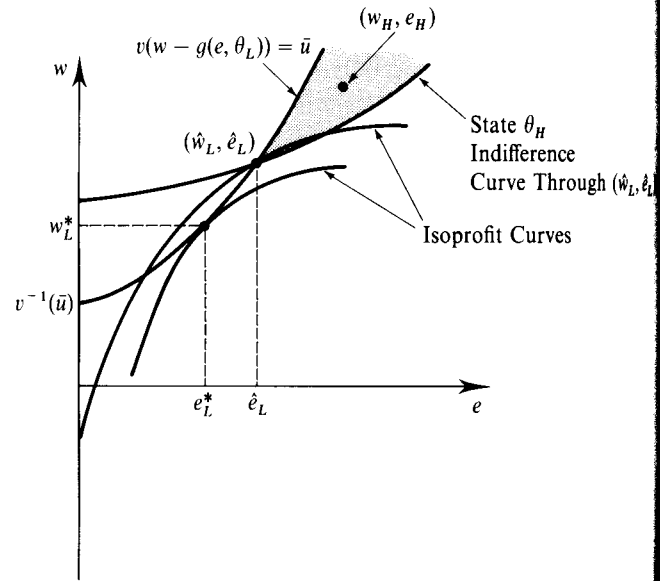
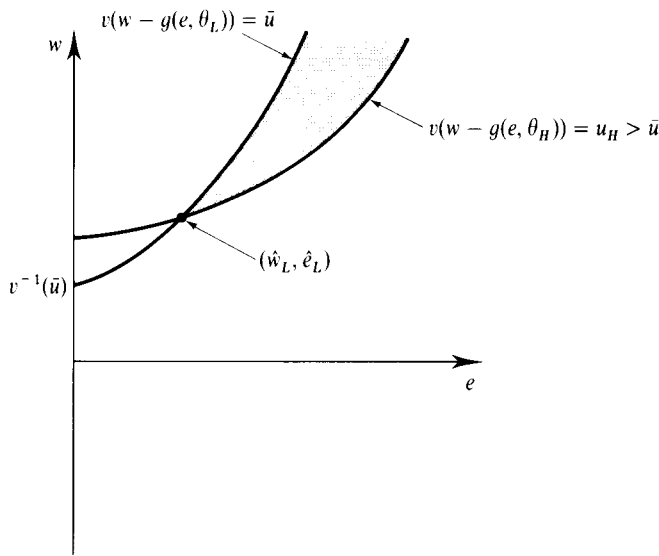
**Lemma 14.C.2:** An optimal contract in problem (14.C.8) must have  $w_L - g(e_L, \theta_L) = v^{-1}(\bar{u})$ .

**Proof:** Suppose not, that is, that there is an optimal solution  $[(w_L, e_L), (w_H, e_H)]$  in which  $w_L - g(e_L, \theta_L) > v^{-1}(\bar{u})$ . Now, consider an alteration to the owner's contract



**Figure 14.C.3**

Constraint (ii) in problem (14.C.8) is satisfied by any contract satisfying constraints (i) and (iii).



in which the owner pays wages in the two states of  $\hat{w}_L = w_L - \varepsilon$  and  $\hat{w}_H = w_H - \varepsilon$ , where  $\varepsilon > 0$  (i.e., the owner lowers the wage payments in both states by  $\varepsilon$ ). This new contract still satisfies constraint (i) as long as  $\varepsilon$  is chosen small enough. In addition, the incentive compatibility constraints are still satisfied because this change just subtracts a constant,  $\varepsilon$ , from each side of these constraints. But if this new contract satisfies all the constraints, the original contract could not have been optimal because the owner now has higher profits, which is a contradiction. ■

**Figure 14.C.4 (left)**

In a feasible contract offering  $(\hat{w}_L, \hat{e}_L)$  for state  $\theta_L$ , the pair  $(w_H, e_H)$  must lie in the shaded region.

**Figure 14.C.5 (right)**

An optimal contract has  $e_L \leq e_L^*$ .

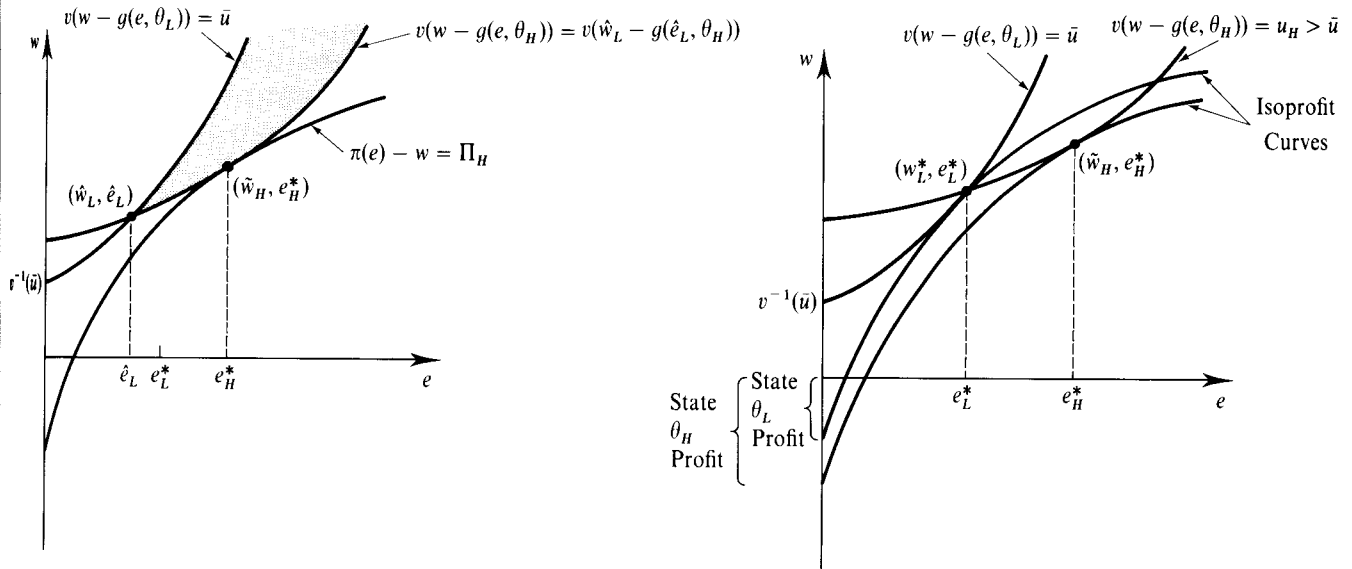
**Lemma 14.C.3:** In any optimal contract:

- (i)  $e_L \leq e_L^*$ ; that is, the manager's effort level in state  $\theta_L$  is no more than the level that would arise if  $\theta$  were observable.
- (ii)  $e_H = e_H^*$ ; that is, the manager's effort level in state  $\theta_H$  is exactly equal to the level that would arise if  $\theta$  were observable.

**Proof:** Lemma 14.C.3 can best be seen graphically. By Lemma 14.C.2,  $(w_L, e_L)$  lies on the locus  $\{(w, e): v(w - g(e, \theta_L)) = \bar{u}\}$  in any optimal contract. Figure 14.C.4 depicts one possible pair  $(\hat{w}_L, \hat{e}_L)$ . In addition, the truth-telling constraints imply that the outcome for state  $\theta_H$ ,  $(w_H, e_H)$ , must lie in the shaded region of Figure 14.C.4. To see this, note that by constraint (iv),  $(w_H, e_H)$  must lie on or below the state  $\theta_L$  indifference curve through  $(\hat{w}_L, \hat{e}_L)$ . In addition, by constraint (iii),  $(w_H, e_H)$  must lie on or above the state  $\theta_H$  indifference curve through  $(\hat{w}_L, \hat{e}_L)$ .

To see part (i), suppose that we have a contract with  $\hat{e}_L > e_L^*$ . Figure 14.C.5 depicts such a contract offer:  $(\hat{w}_L, \hat{e}_L)$  lies on the manager's state  $\theta_L$  indifference curve with utility level  $\bar{u}$ , and  $(w_H, e_H)$  lies in the shaded region defined by the truth-telling constraints. The state  $\theta_L$  indifference curve for the manager and the isoprofit curve for the owner which go through point  $(\hat{w}_L, \hat{e}_L)$  have the relation depicted at point  $(\hat{w}_L, \hat{e}_L)$  because  $\hat{e}_L > e_L^*$ .

As can be seen in the figure, the owner can raise his profit level in state  $\theta_L$  by moving the state  $\theta_L$  wage-effort pair down the manager's indifference curve from  $(\hat{w}_L, \hat{e}_L)$  to its first-best point  $(w_L^*, e_L^*)$ . This change continues to satisfy all the constraints in problem (14.C.8): The manager's utility in each state is unchanged,

**Figure 14.C.6 (left)**

An optimal contract has  $e_H = e_H^*$ .

**Figure 14.C.7 (right)**

The best contract with  $e_L = e_L^*$ .

and, as is evident in Figure 14.C.5, the truth-telling constraints are still satisfied. Thus, a contract with  $\hat{e}_L > e_L^*$  cannot be optimal.

Now consider part (ii). Given any wage–effort pair  $(\hat{w}_L, \hat{e}_L)$  with  $\hat{e}_L \leq e_L^*$ , such as that shown in Figure 14.C.6, the owner's problem is to find the location for  $(w_H, e_H)$  in the shaded region that maximizes his profit in state  $\theta_H$ . The solution occurs at a point of tangency between the manager's state  $\theta_H$  indifference curve through point  $(\hat{w}_L, \hat{e}_L)$  and an isoprofit curve for the owner. This tangency occurs at point  $(\tilde{w}_H, e_H^*)$  in the figure, and necessarily involves effort level  $e_H^*$  because all points of tangency between the manager's state  $\theta_H$  indifference curves and the owner's isoprofit curves occur at effort level  $e_H^*$  [they are characterized by condition (14.C.7) for  $i = H$ ]. Note that this point of tangency occurs strictly to the right of effort level  $\hat{e}_L$  because  $\hat{e}_L \leq e_L^* < e_H^*$ . ■

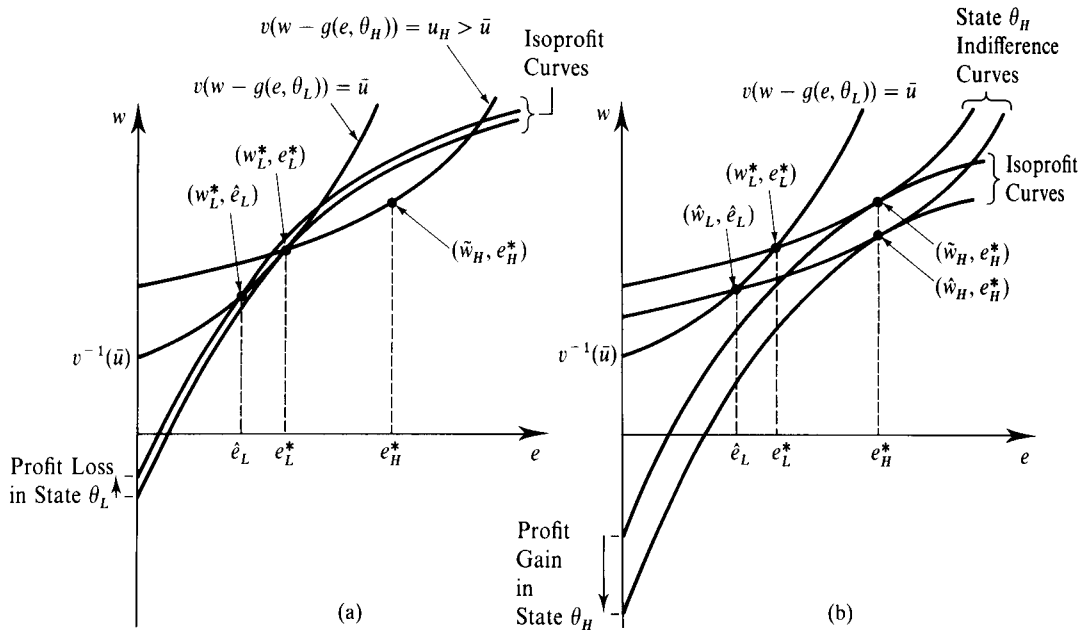
A secondary point emerging from the proof of Lemma 14.C.3 is that only the truth-telling constraint for state  $\theta_H$  is binding in the optimal contract. This property is common to many of the other applications in the literature.<sup>16</sup>

**Lemma 14.C.4:** In any optimal contract,  $e_L < e_L^*$ ; that is, the effort level in state  $\theta_L$  is necessarily *strictly* below the level that would arise in state  $\theta_L$  if  $\theta$  were observable.

**Proof:** Again, this point can be seen graphically. Suppose we start with  $(w_L, e_L) = (w_L^*, e_L^*)$ , as in Figure 14.C.7. By Lemma 14.C.3, this determines the state  $\theta_H$  outcome, denoted by  $(\tilde{w}_H, e_H^*)$  in the figure. Note that by the definition of  $(w_L^*, e_L^*)$ , the isoprofit curve through this point is tangent to the manager's state  $\theta_L$  indifference curve.

Recall that the absolute distance between the origin and the point where each state's isoprofit curve hits the vertical axis represents the profit the owner earns in that state. The owner's overall expected profit with this contract offer is therefore

16. In models with more than two types, this property takes the form that only the incentive constraints between adjacent types bind, and they do so only in one direction. (See Exercise 14.C.1.)



**Figure 14.C.8** (a) The change in profits in state  $\theta_L$  from lowering  $e_L$  slightly below  $e_L^*$ . (b) The change in profits in state  $\theta_L$  from lowering  $e_L$  slightly below  $e_L^*$  and optimally adjusting  $w_H$ .

equal to the average of these two profit levels (with weights equal to the relative probabilities of the two states).

We now argue that a change in the state  $\theta_L$  outcome that lowers this state's effort level to one slightly below  $e_L^*$  necessarily raises the owner's expected profit. To see this, start by moving the state  $\theta_L$  outcome to a slightly lower point,  $(\hat{w}_L, \hat{e}_L)$ , on the manager's state  $\theta_L$  indifference curve. This change is illustrated in Figure 14.C.8, along with the owner's isoprofit curve through this new point. As is evident in Figure 14.C.8(a), this change lowers the profit that the owner earns in state  $\theta_L$ . However, it also relaxes the incentive constraint on the state  $\theta_H$  outcome and, by doing so, it allows the owner to offer a lower wage in that state. Figure 14.C.8(b) shows the new state  $\theta_H$  outcome, say  $(\hat{w}_H, e_H^*)$ , and the new (higher-profit) isoprofit curve through this point.

Overall, this change results in a lower profit for the owner in state  $\theta_L$  and a higher profit for the owner in state  $\theta_H$ . Note, however, that because we started at a point of tangency at  $(w_L^*, e_L^*)$ , the profit loss in state  $\theta_L$  is small relative to the gain in state  $\theta_H$ . Indeed, if we were to look at the derivative of the owner's profit in state  $\theta_L$  with respect to an *infinitesimal* change in that state's outcome, we would find that it is *zero*. In contrast, the derivative of profit in state  $\theta_H$  with respect to this infinitesimal change would be strictly positive. The zero derivative in state  $\theta_L$  is an envelope theorem result: because we started out at the first-best level of effort in state  $\theta_L$ , a small change in  $(w_L, e_L)$  that keeps the manager's state  $\theta_L$  utility at  $\bar{u}$  has no first-order effect on the owner's profit in that state; but because it relaxes the state  $\theta_H$  incentive constraint, for a small-enough change the owner's expected profit is increased. ■

How far should the owner go in lowering  $e$ ? In answering this question, the owner must weigh the marginal loss in profit in state  $\theta_L$  against the marginal gain in state

$\theta_H$  [note that once we move away from  $(w_L^*, e_L^*)$ , the envelope result no longer holds and the marginal reduction in state  $\theta_L$ 's profit is strictly positive]. It should not be surprising that the extent to which the owner wants to make this trade-off depends on the relative probabilities of the two states. In particular, the greater the likelihood of state  $\theta_H$ , the more the owner is willing to distort the state  $\theta_L$  outcome to increase profit in state  $\theta_H$ . In the extreme case in which the probability of state  $\theta_L$  gets close to zero, the owner may set  $e_L = 0$  and hire the manager to work only in state  $\theta_H$ .<sup>17</sup>

The analysis in Appendix B confirms this intuition. There we show that the optimal level of  $e_L$  satisfies the following first-order condition:

$$[\pi'(e_L) - g_e(e_L, \theta_L)] + \frac{\lambda}{1 - \lambda} [g_e(e_L, \theta_H) - g_e(e_L, \theta_L)] = 0. \quad (14.C.9)$$

The first term of this expression is zero at  $e_L = e_L^*$  and is strictly positive at  $e_L < e_L^*$ ; the second term is always strictly negative. Thus, we must have  $e_L < e_L^*$  to satisfy this condition, confirming our finding in Lemma 14.C.4. Differentiating this expression reveals that the optimal level of  $e_L$  falls as  $\lambda/(1 - \lambda)$  rises.

These findings are summarized in Proposition 14.C.3.

**Proposition 14.C.3:** In the hidden information principal-agent model with an infinitely risk-averse manager the optimal contract sets the level of effort in state  $\theta_H$  at its first-best (full observability) level  $e_H^*$ . The effort level in state  $\theta_L$  is distorted downward from its first-best level  $e_L^*$ . In addition, the manager is inefficiently insured, receiving a utility greater than  $\bar{u}$  in state  $\theta_H$  and a utility equal to  $\bar{u}$  in state  $\theta_L$ . The owner's expected payoff is strictly lower than the expected payoff he receives when  $\theta$  is observable, while the infinitely risk-averse manager's expected utility is the same as when  $\theta$  is observable (it equals  $\bar{u}$ ).<sup>18,19</sup>

A basic, and very general, point that emerges from this analysis is that the optimal contract for the owner in this setting of hidden information necessarily *distorts* the effort choice of the manager in order to ameliorate the costs of asymmetric information, which here take the form of the higher expected wage payment that the owner makes because the manager has a utility in state  $\theta_H$  in excess of  $\bar{u}$ .

Note that nothing would change if the profit level  $\pi$  were not publicly observable (and so could not be contracted on), since our analysis relied only on the fact that the effort level  $e$  was observable. Moreover, in the case in which  $\pi$  is not publicly observable, we can extend the model to allow the relationship between profits and effort to depend on the state; that is, the owner's profits in states  $\theta_L$  and  $\theta_H$  given effort level  $e$  might be given by the functions  $\pi_L(e)$  and  $\pi_H(e)$ .<sup>20</sup> As long as

17. In fact, this can happen only if  $g_e(0, \theta_L) > 0$ .

18. Recall that an infinitely risk-averse manager's expected utility is equal to his lowest utility level across the two states.

19. Note, however, that while the outcome here is Pareto inefficient, it is a constrained Pareto optimum in the sense introduced in Section 13.B; the reasons parallel those given in footnote 9 of Section 14.B for the hidden action model (although here it is  $\theta$  that the authority cannot observe rather than  $e$ ).

20. The nonobservability of profits is important for this extension because if  $\pi$  could be contracted upon, the manager could be punished for misrepresenting the state by simply comparing the realized profit level with the profit level that should have been realized in the announced state for the specified level of effort.

$\pi'_H(e) \geq \pi'_L(e) > 0$  for all  $e \geq 0$ , the analysis of this model follows exactly along the lines of the analysis we have just conducted (see Exercise 14.C.5).

As in the case of hidden action models, a number of extensions of this basic hidden information model have been explored in the literature. Some of the most general treatments appear in the context of the “mechanism design” literature associated with social choice theory. A discussion of these models can be found in Chapter 23.

### *The Monopolistic Screening Model*

In Section 13.D, we studied a model of *competitive screening* in which firms try to design their employment contracts in a manner that distinguishes among workers who, at the time of contracting, have different unobservable productivity levels (i.e., there is *precontractual* asymmetric information). The techniques that we have developed in our study of the principal-agent model with hidden information enable us to formulate and solve a model of *monopolistic screening* in which, in contrast with the analysis in Section 13.D, only a single firm offers employment contracts (actually, this might more properly be called a *monopsonistic* screening model because the single firm is on the demand side of the market).

To see this, suppose that, as in the model in Section 13.D, there are two possible types of workers who differ in their productivity. A worker of type  $\theta$  has utility  $u(w, t | \theta) = w - g(t, \theta)$  when he receives a wage of  $w$  and faces task level  $t$ . His reservation utility level is  $\bar{u}$ . The productivities of the two types of workers are  $\theta_H$  and  $\theta_L$ , with  $\theta_H > \theta_L > 0$ . The fraction of workers of type  $\theta_H$  is  $\lambda \in (0, 1)$ . We assume that the firm's profits, which are not publicly observable, are given by the function  $\pi_H(t)$  for a type  $\theta_H$  worker and by  $\pi_L(t)$  for a type  $\theta_L$  worker, and that  $\pi'_H(t) \geq \pi'_L(t) > 0$  for all  $t \geq 0$  [e.g., as in Exercise 13.D.1, we could have  $\pi_i(t) = \theta_i(1 - \mu t)$  for  $\mu > 0$ ].<sup>21</sup>

The firm's problem is to offer a set of contracts that maximizes its profits given worker self-selection among, and behavior within, its offered contracts. Once again, the revelation principle can be invoked to greatly simplify the firm's problem. Here the firm can restrict its attention to offering a menu of wage–task pairs  $[(w_H, t_H), (w_L, t_L)]$  to solve

$$\text{Max}_{w_H, t_H \geq 0, w_L, t_L \geq 0} \quad \lambda[\pi_H(t_H) - w_H] + (1 - \lambda)[\pi_L(t_L) - w_L] \quad (14.C.10)$$

- s.t. (i)  $w_L - g(t_L, \theta_L) \geq \bar{u}$   
(ii)  $w_H - g(t_H, \theta_H) \geq \bar{u}$   
(iii)  $w_H - g(t_H, \theta_H) \geq w_L - g(t_L, \theta_H)$   
(iv)  $w_L - g(t_L, \theta_L) \geq w_H - g(t_H, \theta_L)$ .

This problem has exactly the same structure as (14.C.8) but with the principal's (here the firm's) profit being a function of the state. As noted above, the analysis of this problem follows exactly the same lines as our analysis of problem (14.C.8).

This class of models has seen wide application in the literature (although often with a continuum of types assumed). Maskin and Riley (1984b), for example, apply this model to the study of monopolistic price discrimination. In their model, a consumer of type  $\theta$  has utility  $v(x, \theta) - T$  when he consumes  $x$  units of a monopolist's good and makes a total payment of  $T$  to the monopolist, and can earn a reservation utility level of  $v(0, \theta) = 0$  by not purchasing from the monopolist. The monopolist has a constant unit cost of production equal to  $c > 0$

21. The model studied in Section 13.D with  $\pi_i(t) = \theta_i$  corresponds to the limiting case where  $\mu \rightarrow 0$ .

and seeks to offer a menu of  $(x_i, T_i)$  pairs to maximize its profit. The monopolist's problem then takes the form in (14.C.10) where we take  $t_i = x_i$ ,  $w_i = -T_i$ ,  $\bar{u} = 0$ ,  $g(t_i, \theta_i) = -v(x_i, \theta_i)$ , and  $\pi_i(t_i) = -cx_i$ .

Baron and Myerson's (1982) analysis of optimal regulation of a monopolist with unknown costs provides another example. There, a regulated firm faces market demand function  $x(p)$  and has unobservable unit costs of  $\theta$ . The regulator, who seeks to design a regulatory policy that maximizes consumer surplus, faces the monopolist with a choice among a set of pairs  $(p_i, T_i)$ , where  $p_i$  is the allowed retail price and  $T_i$  is a transfer payment from the regulator to the firm. The regulated firm is able to shut down if it cannot earn profits of at least zero from any of the regulator's offerings. The regulator's problem then corresponds to (14.C.10) with  $t_i = p_i$ ,  $w_i = T_i$ ,  $\bar{u} = 0$ ,  $g(t_i, \theta_i) = -(p_i - \theta_i)x(p_i)$ , and  $\pi_i(t_i) = \int_{p_i}^{\infty} x(s) ds$ .<sup>22</sup>

Exercises 14.C.7 to 14.C.9 ask you to study some examples of monopolistic screening models.

## 14.D Hidden Actions and Hidden Information: Hybrid Models

Although the hidden action–hidden information dichotomization serves as a useful starting point for understanding principal-agent models, many real-world situations (and some of the literature as well) involve elements of both problems.

To consider an example of such a model, suppose that we augment the simple hidden information model considered in Section 14.C in the following manner: let the level of effort  $e$  now be unobservable, and let profits be a stochastic function of effort, described by conditional density function  $f(\pi|e)$ . In essence, what we now have is a hidden action model, but one in which the owner also does not know something about the disutility of the manager (which is captured in the state variable  $\theta$ ).

Formal analysis of this model is beyond the scope of this chapter, but the basic thrust of the revelation principle extends to the analysis of these types of hybrid problems. In particular, as Myerson (1982) shows, the owner can now restrict attention to contracts of the following form:

- (i) After the state  $\theta$  is realized, the manager announces which state has occurred.
- (ii) The contract specifies, for each possible announcement  $\hat{\theta} \in \Theta$ , the effort level  $e(\hat{\theta})$  that the manager should take and a compensation scheme  $w(\pi|\hat{\theta})$ .
- (iii) In every state  $\theta$ , the manager is willing to be both *truthful* in stage (i) and *obedient* following stage (ii) [i.e., he finds it optimal to choose effort level  $e(\theta)$  in state  $\theta$ ].

This contract can be thought of as a revelation game, but one in which the outcome of the manager's announcement about the state is a hidden action-style contract, that is, a compensation scheme and a "recommended action." The requirement of "obedience" amounts to an incentive constraint that is like that in the hidden action

22. The regulator's objective function can be generalized to allow a weighted average of consumer and producer surplus, with greater weight on consumers. In this case, the function  $\pi_i(\cdot)$  will depend on  $\theta_i$ .

model considered in Section 14.B; the “truthfulness” constraints are generalizations of those considered in our hidden information model. See Myerson (1982) for details.

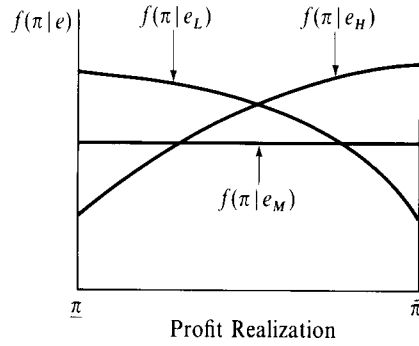
One special case of this hybrid model deserves particular mention because its analysis reduces to that of the pure hidden information model considered in Section 14.C. In particular, suppose that effort is unobservable but that the relationship between effort and profits is *deterministic*, given by the function  $\pi(e)$ . In that case, for any particular announcement  $\hat{\theta}$ , it is possible to induce any wage–effort pair that is desired, say  $(\hat{w}, \hat{e})$ , by use of a simple “forcing” compensation scheme: Just reward the manager with a wage payment of  $\hat{w}$  if profits are  $\pi(\hat{e})$ , and give him a wage payment of  $-\infty$  otherwise. Thus, the combination of the observability of  $\pi$  and the one-to-one relationship between  $\pi$  and  $e$  effectively allows the contract to specify  $e$ . The analysis of this model is therefore identical to that of the hidden information model considered in Section 14.C, where wage–effort pairs could be specified directly as functions of the manager’s announcement.

To see this point in a slightly different way, note first that because of the ability to write forcing contracts, in this model an optimal contract can be thought of as specifying, for each announcement  $\hat{\theta}$ , a wage–profit pair  $(w(\hat{\theta}), \pi(\hat{\theta}))$ . Now, for any required profit level  $\pi$ , the effort level necessary to achieve a profit of  $\pi$  is  $\tilde{e}$  such that  $\pi(\tilde{e}) = \pi$ . Let the function  $\tilde{e}(\pi)$  describe this effort level. We can now think of the manager as having a disutility function defined directly over the profit level which is given by  $\tilde{g}(\pi, \theta) = g(\tilde{e}(\pi), \theta)$ . But this model looks just like a model with *observable* effort where the effort variable is  $\pi$ , the disutility function over this effort is  $\tilde{g}(\pi, \theta)$ , and the profit function is  $\tilde{\pi}(\pi) = \pi$ . Thus, the analysis of this model is identical to that in a pure hidden information model.

A similar point applies to a closely related hybrid model in which, instead of the manager’s disutility of effort, it is the relation between profit and effort that depends on the state. In particular, suppose that the disutility of effort is given by the function  $g(e)$  and profits are given by the function  $\pi(e, \theta)$ , where  $\pi_e(\cdot) > 0$ ,  $\pi_{ee}(\cdot) < 0$ ,  $\pi_\theta(\cdot) > 0$ , and  $\pi_{e\theta}(\cdot) > 0$ . Effort is not observable, but profits are. The idea is that the manager knows more than the owner does about the true profit opportunities facing the firm (e.g., the marginal productivity of effort). Again, we can think of a contract as specifying, for each announcement by the manager, a wage–profit pair (implicitly using forcing contracts). In this context, the effort needed to achieve any given level of profit  $\pi$  in state  $\theta$  is given by some function  $\hat{e}(\pi, \theta)$ , and the disutility associated with this effort is then  $\hat{g}(\pi, \theta) = g(\hat{e}(\pi, \theta))$ . But this model is also equivalent to our basic hidden information model with observable effort: just let the effort variable be  $\pi$ , the disutility of this effort be  $\hat{g}(\pi, \theta)$ , and the profit function be  $\hat{\pi}(\pi) = \pi$ . Again, our results from Section 14.C apply.

## APPENDIX A: MULTIPLE EFFORT LEVELS IN THE HIDDEN ACTION MODEL

In this appendix, we discuss additional issues that arise when the effort choice in the hidden action (moral hazard) model discussed in Section 14.B is more complex than the simple two-effort-choice specification  $e \in \{e_L, e_H\}$  analyzed there. Here, we return


**Figure 14.AA.1**

Density functions for  $E = \{e_L, e_M, e_H\}$ : effort choice  $e_M$  may not be implementable.

to the more general specification initially introduced in Section 14.B in which  $E$  is the feasible set of effort choices.

As in Section 14.B, we can break up the principal's (the owner's) problem into several parts:

- What are the effort levels  $e$  that it is possible to induce?
- What is the optimal contract for inducing each specific effort level  $e \in E$ ?
- Which effort level  $e \in E$  is optimal?

In a multiple-action setting, each of these three parts becomes somewhat more complicated. For example, with just two actions, part (a) was trivial:  $e_L$  could be induced with a fixed wage contract, and  $e_H$  could always be induced by giving incentives that were sufficiently high at outcomes that were more likely to arise when  $e_H$  is chosen. With more than two actions, however, this may not be so. For example, consider the three-action case in which  $E = \{e_L, e_M, e_H\}$  and the conditional density functions are those depicted in Figure 14.AA.1. As is suggested by the figure, it may be impossible to design incentives such that  $e_M$  is chosen because for any  $w(\pi)$  the agent may prefer either  $e_L$  or  $e_H$  to  $e_M$ . (Exercise 14.B.4 provides an example along these lines.)

Part (b) also becomes more involved. The optimal contract for implementing effort choice  $e$  solves

$$\begin{aligned} \text{Min}_{w(\pi)} \quad & \int w(\pi) f(\pi|e) d\pi \\ \text{s.t. (i)} \quad & \int v(w(\pi)) f(\pi|e) d\pi - g(e) \geq \bar{u} \\ \text{(ii)} \quad & e \text{ solves } \text{Max}_{\tilde{e} \in E} \int v(w(\pi)) f(\pi|\tilde{e}) d\pi - g(\tilde{e}). \end{aligned} \tag{14.AA.1}$$

If we have  $K$  possible actions in set  $E$ , the incentive constraints in problem (14.AA.1) [constraints (ii)] consist of  $(K - 1)$  constraints that must be satisfied. In this case, with a change of variables in which we maximize over the level of utility that the manager gets conditional on  $\pi$ , say  $\bar{v}(\pi)$ , we have a problem with  $K$  linear constraints and a convex objective function [see Grossman and Hart (1983) and footnote 7 for more on this].

However, if  $E$  is a continuous set of possible actions, say  $E = [0, \bar{e}] \subset \mathbb{R}$ , then we have an *infinity* of incentive constraints. One trick sometimes used in this case to

simplify problem (14.AA.1) is to replace constraint (ii) with a *first-order condition* (this is sometimes called the *first-order approach*). For example, if  $e$  is a one-dimensional measure of effort, then the manager's first-order condition is

$$\int v(w(\pi)) f_e(\pi|e) d\pi - g'(e) = 0, \quad (14.AA.2)$$

where  $f_e(\pi|e) = \partial f(\pi|e)/\partial e$ . If we replace constraint (ii) with (14.AA.2) and solve the resulting problem, we can derive a condition for  $w(\pi)$  that parallels condition (14.B.10):

$$\frac{1}{v'(w(\pi))} = \lambda + \mu \left[ \frac{f_e(\pi|e)}{f(\pi|e)} \right]. \quad (14.AA.3)$$

The condition that ratio  $[f_e(\pi|e)/f(\pi|e)]$  be increasing in  $\pi$  is the differential version of the monotone likelihood ratio property (see Exercise 14.AA.1).

In general, however, a solution to the problem resulting from this substitution is not necessarily a solution to the actual problem (14.AA.1). The reason is that the agent may satisfy first-order condition (14.AA.2) even when effort level  $e$  is not his optimal effort choice. First, effort level  $e$  could be a *minimum* rather than a maximum; therefore, we at least want the agent to also be satisfying a local second-order condition. But even this will not be sufficient. In general, we need to be sure that the agent's objective function is concave in  $e$ . Note that this is not a simple matter because the concavity of his objective function in  $e$  will depend both on the shape of  $f(\pi|e)$  and on the shape of the incentive contract  $w(\pi)$  that is offered. The known conditions which insure that this condition is met are very restrictive. See Grossman and Hart (1983) and Rogerson (1985b) for details. Exercise 14.AA.2 provides a very simple example.

Finally, to answer part (c), we need to compute the optimal contract from part (b) for each action that part (a) reveals is implementable and then compare their relative profits for the principal. With more than two effort choices, two features of the two-effort-choice case fail to generalize. First, nonobservability can lead to an upward distortion in effort. (Exercise 14.B.4 provides an example.) Second, at the optimal contract under nonobservability we can get *both* an inefficient effort choice and inefficiencies resulting from managerial risk bearing.

## APPENDIX B: A FORMAL SOLUTION OF THE PRINCIPAL-AGENT PROBLEM WITH HIDDEN INFORMATION

Recall problem (14.C.8):

$$\begin{aligned} & \text{Max}_{w_H, e_H \geq 0, w_L, e_L \geq 0} \quad \lambda[\pi(e_H) - w_H] + (1 - \lambda)[\pi(e_L) - w_L] \\ & \text{s.t.} \quad \text{(i)} \quad w_L - g(e_L, \theta_L) \geq v^{-1}(\bar{u}) \\ & \quad \quad \text{(ii)} \quad w_H - g(e_H, \theta_H) \geq v^{-1}(\bar{u}) \\ & \quad \quad \text{(iii)} \quad w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \\ & \quad \quad \text{(iv)} \quad w_L - g(e_L, \theta_L) \geq w_H - g(e_H, \theta_L). \end{aligned}$$

Using Lemma 14.C.1 we can restate problem (14.C.8) as

$$\text{Max}_{w_H, e_H \geq 0, w_L, e_L \geq 0} \lambda[\pi(e_H) - w_H] + (1 - \lambda)[\pi(e_L) - w_L] \quad (14.BB.1)$$

$$\text{s.t. (i) } w_L - g(e_L, \theta_L) \geq v^{-1}(\bar{u})$$

$$\text{(iii) } w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_L)$$

$$\text{(iv) } w_L - g(e_L, \theta_L) \geq w_H - g(e_H, \theta_H).$$

Letting  $(\gamma, \phi_H, \phi_L) \geq 0$  be the multipliers on constraints (i), (iii), and (iv), respectively, the Kuhn–Tucker conditions for this problem can be written (see Section M.K of the Mathematical Appendix)

$$-\lambda + \phi_H - \phi_L = 0. \quad (14.BB.2)$$

$$-(1 - \lambda) + \gamma - \phi_H + \phi_L = 0. \quad (14.BB.3)$$

$$\lambda \pi'(e_H) - \phi_H g_e(e_H, \theta_H) + \phi_L g_e(e_H, \theta_L) \begin{cases} \leq 0 \\ = 0 \end{cases} \quad \text{if } e_H > 0 \quad (14.BB.4)$$

$$(1 - \lambda) \pi'(e_L) - (\gamma + \phi_L) g_e(e_L, \theta_L) + \phi_H g_e(e_L, \theta_H) \begin{cases} \leq 0 \\ = 0 \end{cases} \quad \text{if } e_L > 0 \quad (14.BB.5)$$

along with the complementary slackness conditions for constraints (i), (iii), and (iv) [conditions (M.K.7)].

Let us break up the analysis of these conditions into several steps.

*Step 1:* Condition (14.BB.2) implies that  $\phi_H > 0$ . Thus, constraint (iii) must bind (hold with equality) at an optimal solution.

*Step 2:* Adding conditions (14.BB.2) and (14.BB.3) implies that  $\gamma = 1$ . Hence, constraint (i) must bind at an optimal solution.

*Step 3:* Both  $e_L$  and  $e_H$  are strictly positive. To see this, note that condition (14.BB.4) cannot hold at  $e_H = 0$  because  $\pi'(0) > 0$  and  $g_e(0, \theta_i) = 0$  for  $i = L, H$ . Similarly for condition (14.BB.5) and  $e_L$ .

*Step 4:* Steps 1 to 3 imply that  $\phi_L = 0$ . Suppose not: i.e., that  $\phi_L > 0$ . Then constraint (iv) must be binding. We shall now derive a contradiction. First, substitute for  $\phi_H$  in conditions (14.BB.4) and (14.BB.5) using the fact that  $\phi_H = \phi_L + \lambda$  from condition (14.BB.2). Then, using the fact that  $(e_L, e_H) \gg 0$ , we can write conditions (14.BB.4) and (14.BB.5) as

$$\lambda[\pi'(e_H) - g_e(e_H, \theta_H)] + \phi_L[g_e(e_H, \theta_L) - g_e(e_H, \theta_H)] = 0$$

and

$$(1 - \lambda)[\pi'(e_L) - g_e(e_L, \theta_H)] + (1 + \phi_L)[g_e(e_L, \theta_H) - g_e(e_L, \theta_L)] = 0.$$

But  $\phi_L > 0$  then implies that

$$\pi'(e_L) - g_e(e_L, \theta_H) > 0 > \pi'(e_H) - g_e(e_H, \theta_H),$$

which implies  $e_H > e_L$  since  $\pi(e) - g(e, \theta_H)$  is concave in  $e$ . But if  $e_H > e_L$  and constraint (iii) binds (which it does from Step 1), then constraint (iv) must be slack

because we then have

$$\begin{aligned}
 (w_H - w_L) &= g(e_H, \theta_H) - g(e_L, \theta_H) \\
 &= \int_{e_L}^{e_H} g_e(e, \theta_H) de \\
 &< \int_{e_L}^{e_H} g_e(e, \theta_L) de \\
 &= g(e_H, \theta_L) - g(e_L, \theta_L).
 \end{aligned}$$

This is our desired contradiction.

*Step 5:* Since  $\phi_L = 0$ , we know from (14.BB.2) that  $\phi_H = \lambda$ . Substituting these two values into conditions (14.BB.4) and (14.BB.5) we have

$$\pi'(e_H) - g_e(e_H, \theta_H) = 0 \quad (14.BB.6)$$

and

$$[\pi'(e_L) - g_e(e_L, \theta_L)] + \frac{\lambda}{1 - \lambda} [g_e(e_L, \theta_H) - g_e(e_L, \theta_L)] = 0. \quad (14.BB.7)$$

Conditions (14.BB.6) and (14.BB.7) characterize the optimal values of  $e_H$  and  $e_L$ , respectively. The optimal values for  $w_L$  and  $w_H$  are then determined from constraints (i) and (iii), which we have seen hold with equality at the solution.

An alternative approach to solving problem (14.BB.1) that avoids this somewhat cumbersome argument involves the following “trick”: Solve problem (14.BB.1) ignoring constraint (iv). Then show that the solution derived in this way also satisfies constraint (iv). If so, this must be a solution to the (more constrained) problem (14.BB.1). (Exercise 14.BB.1 asks you to try this approach.)

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## EXERCISES

**14.B.1<sup>B</sup>** Consider the two-effort-level hidden action model discussed in Section 14.B with the general utility function  $u(w, e)$  for the agent. Must the reservation utility constraint be binding in an optimal contract?

**14.B.2<sup>B</sup>** Derive the first-order condition characterizing the optimal compensation scheme for the two-effort-level hidden action model studied in Section 14.B when the principal is strictly risk averse.

**14.B.3<sup>B</sup>** Consider a hidden action model in which the owner is risk neutral while the manager has preferences defined over the mean and the variance of his income  $w$  and his effort level  $e$  as follows: Expected utility =  $E[w] - \phi \text{Var}(w) - g(e)$ , where  $g'(0) = 0$ ,  $(g'(e), g''(e), g'''(e)) \gg 0$  for  $e > 0$ , and  $\lim_{e \rightarrow \infty} g'(e) = \infty$ . Possible effort choices are  $e \in \mathbb{R}_+$ . Conditional on effort level  $e$ , the realization of profit is normally distributed with mean  $e$  and variance  $\sigma^2$ .

(a) Restrict attention to linear compensation schemes  $w(\pi) = \alpha + \beta\pi$ . Show that the manager's expected utility given  $w(\pi)$ ,  $e$ , and  $\sigma^2$  is given by  $\alpha + \beta e - \phi\beta^2\sigma^2 - g(e)$ .

(b) Derive the optimal contract when  $e$  is observable.

(c) Derive the optimal linear compensation scheme when  $e$  is not observable. What effects do changes in  $\beta$  and  $\sigma^2$  have?

**14.B.4<sup>B</sup>** Consider the following hidden action model with three possible actions  $E = \{e_1, e_2, e_3\}$ . There are two possible profit outcomes:  $\pi_H = 10$  and  $\pi_L = 0$ . The probabilities of  $\pi_H$  conditional on the three effort levels are  $f(\pi_H|e_1) = \frac{2}{3}$ ,  $f(\pi_H|e_2) = \frac{1}{2}$ , and  $f(\pi_H|e_3) = \frac{1}{3}$ . The agent's effort cost function has  $g(e_1) = \frac{5}{3}$ ,  $g(e_2) = \frac{8}{3}$ ,  $g(e_3) = \frac{4}{3}$ . Finally,  $v(w) = \sqrt{w}$ , and the manager's reservation utility is  $\bar{u} = 0$ .

(a) What is the optimal contract when effort is observable?

(b) Show that if effort is not observable, then  $e_2$  is not implementable. For what levels of  $g(e_2)$  would  $e_2$  be implementable? [Hint: Focus on the utility levels the manager will get for the two outcomes,  $v_1$  and  $v_2$ , rather than on the wage payments themselves.]

(c) What is the optimal contract when effort is not observable?

(d) Suppose, instead, that  $g(e_1) = \sqrt{8}$ , and let  $f(\pi_H|e_1) = x \in (0, 1)$ . What is the optimal contract if effort is observable as  $x$  approaches 1? What is the optimal contract as  $x$  approaches 1 if it is not observable? As  $x$  approaches 1, is the level of effort implemented higher or lower when effort is not observable than when it is observable?

**14.B.5<sup>B</sup>** Suppose that in the hidden action model explored in Section 14.B the manager can not only choose how much effort to exert but can also, after observing the realization of the firm's profits  $\pi$ , unobservably reduce them in a way that is of no direct benefit to him (e.g., he can voluntarily offer to pay more for his inputs). Show that in this case there is always an optimal incentive scheme that is nondecreasing in observed profits.

**14.B.6<sup>B</sup>** Amend the two-effort-level model studied in Section 14.B as follows: Suppose now that effort has distinct effects on revenues  $R$  and costs  $C$ , where  $\pi = R - C$ . Let  $f_R(R|e)$  and  $f_C(C|e)$  denote the density functions of  $R$  and  $C$  conditional on  $e$ , and assume that, conditional on  $e$ ,  $R$  and  $C$  are independently distributed. Assume  $R \in [\underline{R}, \bar{R}]$ ,  $C \in [\underline{C}, \bar{C}]$ , and that for all  $e$ ,  $f_R(R|e) > 0$  for all  $R \in [\underline{R}, \bar{R}]$  and  $f_C(C|e) > 0$  for all  $C \in [\underline{C}, \bar{C}]$ .

The two effort choices are now  $\{e_R, e_C\}$ , where  $e_R$  is an effort choice that devotes more time to revenue enhancement and less to cost reduction, and the opposite is true for  $e_C$ . In particular, assume that  $F_R(R|e_R) < F_R(R|e_C)$  for all  $R \in (\underline{R}, \bar{R})$  and that  $F_C(C|e_C) > F_C(C|e_R)$  for all  $C \in (\underline{C}, \bar{C})$ . Moreover, assume that the monotone likelihood ratio property holds for each of these variables in the following form:  $[f_R(R|e_R)/f_R(R|e_C)]$  is increasing in  $R$ , and  $[f_C(C|e_R)/f_C(C|e_C)]$  is increasing in  $C$ . Finally, the manager prefers revenue enhancement over cost reduction: that is,  $g(e_C) > g(e_R)$ .

(a) Suppose that the owner wants to implement effort choice  $e_C$  and that both  $R$  and  $C$  are observable. Derive the first-order condition for the optimal compensation scheme  $w(R, C)$ . How does it depend on  $R$  and  $C$ ?

(b) How would your answer to (a) change if the manager could always unobservably reduce the revenues of the firm (in a way that is of no direct benefit to him)?

(c) What if, in addition, costs are now unobservable by a court (so that compensation can be made contingent only on revenues)?

**14.B.7<sup>C</sup>** Consider a two-period model that involves two repetitions of the two-effort-level hidden action model studied in Section 14.B. There is no discounting by either the firm or the manager. The manager's expected utility over the two periods is the sum of his two single-period expected utilities  $E[v(w) - g(e)]$ , where  $v'(\cdot) > 0$  and  $v''(\cdot) < 0$ .

Suppose that a contract can be signed ex ante that gives payoffs in each period as a function of performance up until then. Will period 2 wages depend on period 1 profits in the optimal contract?

**14.B.8<sup>C</sup>** Amend the two-effort-choice hidden action model discussed in Section 14.B as follows: Suppose the principal can, for a cost of  $c$ , observe an extra signal  $\hat{y}$  of the agent's effort. Profits  $\pi$  and the signal  $y$  have a joint distribution  $f(\pi, y|e)$  conditional on  $e$ . The decision to investigate the value of  $y$  can be made after observing  $\pi$ .

A contract now specifies a wage schedule  $w(\pi)$  in the event of no investigation, a wage schedule  $w(\pi, y)$  if an investigation occurs, and a probability  $p(\pi)$  of investigation conditional on  $\pi$ . Characterize the optimal contract for implementing effort level  $e_H$ .

**14.C.1<sup>C</sup>** Analyze the extension of the hidden information model discussed in Section 14.C where there are an arbitrary finite number of states  $(\theta_1, \dots, \theta_N)$  where  $\theta_{i+1} > \theta_i$  for all  $i$ .

**14.C.2<sup>B</sup>** Consider the hidden information model in Section 14.C, but now let the manager be risk neutral with utility function  $v(w) = w$ . Show that the owner can do as well when  $\theta$  is unobservable as when it is observable. In particular, show that he can accomplish this with a contract that offers the manager a compensation scheme of the form  $w(\pi) = \pi - \alpha$  and allows him to choose any effort level he wants. Graph this function and the manager's choices in  $(w, e)$ -space. What revelation mechanism would give this same outcome?

**14.C.3<sup>B</sup>** Suppose that in the two-state hidden information model examined in Section 14.C,  $u(w, e, \theta) = v(w) - g(e, \theta)$ .

(a) Characterize the optimal contract under full observability.

(b) Is this contract feasible when the state  $\theta$  is not observable?

**14.C.4<sup>C</sup>** Characterize the solution to the two-state principal-agent model with hidden information when the manager is risk averse, but not infinitely so.

**14.C.5<sup>B</sup>** Confirm that the analysis in Section 14.C could not change if the owner's profits depended on the state and were not publicly observable and if, letting  $\pi_i(e)$  denote the profits in state  $\theta_i$  for  $i = L, H$ ,  $\pi'_H(e) \geq \pi'_L(e) > 0$  for all  $e \geq 0$ . What happens if this condition is not satisfied?

**14.C.6<sup>C</sup>** Reconsider the labor market screening model in Exercise 13.D.1, but now suppose that there is a single employer. Characterize the solution to this firm's screening problem (assume that both types of workers have a reservation utility level of 0). Compare the task levels in this solution with those in the equilibrium of the competitive screening model (assuming an equilibrium exists) that you derived in Exercise 13.D.1.

**14.C.7<sup>B</sup>** (J. Tirole) Assume that there are two types of consumers for a firm's product,  $\theta_H$  and  $\theta_L$ . The proportion of type  $\theta_L$  consumers is  $\lambda$ . A type  $\theta$ 's utility when consuming amount  $x$  of the good and paying a total of  $T$  for it is  $u(x, T) = \theta v(x) - T$ , where

$$v(x) = \frac{1 - (1 - x)^2}{2}.$$

The firm is the sole producer of this good, and its cost of production per unit is  $c > 0$ .

(a) Consider a nondiscriminating monopolist. Derive his optimal pricing policy. Show that he serves both classes of consumers if either  $\theta_L$  or  $\lambda$  is "large enough."

(b) Consider a monopolist who can distinguish the two types (by some characteristic) but can only charge a simple price  $p_i$  to each type  $\theta_i$ . Characterize his optimal prices.

(c) Suppose the monopolist cannot distinguish the types. Derive the optimal two-part tariff (a pricing policy consisting of a lump-sum charge  $F$  plus a linear price per unit purchased of  $p$ ) under the assumption that the monopolist serves both types. Interpret. When will the monopolist serve both types?

(d) Compute the fully optimal nonlinear tariff. How do the quantities purchased by the two types compare with the levels in (a) to (c)?

**14.C.8<sup>B</sup>** Air Shangri-la is the only airline allowed to fly between the islands of Shangri-la and Nirvana. There are two types of passengers, tourist and business. Business travelers are willing to pay more than tourists. The airline, however, cannot tell directly whether a ticket purchaser is a tourist or a business traveler. The two types do differ, though, in how much they are willing to pay to avoid having to purchase their tickets in advance. (Passengers do not like to commit themselves in advance to traveling at a particular time.)

More specifically, the utility levels of each of the two types net of the price of the ticket,  $P$ , for any given amount of time  $W$  prior to the flight that the ticket is purchased are given by

$$\begin{aligned} \text{Business: } & v - \theta_B P - W, \\ \text{Tourist: } & v - \theta_T P - W, \end{aligned}$$

where  $0 < \theta_B < \theta_T$ . (Note that for any given level of  $W$ , the business traveler is willing to pay more for his ticket. Also, the business traveler is willing to pay more for any given reduction in  $W$ .)

The proportion of travelers who are tourists is  $\lambda$ . Assume that the cost of transporting a passenger is  $c$ .

Assume in (a) to (d) that Air Shangri-la wants to carry both types of passengers.

(a) Draw the indifference curves of the two types in  $(P, W)$ -space. Draw the airline's isoprofit curves. Now formulate the optimal (profit-maximizing) price discrimination problem mathematically that Air Shangri-la would want to solve. [Hint: Impose nonnegativity of prices as a constraint since, if it charged a negative price, it would sell an infinite number of tickets at this price.]

(b) Show that in the optimal solution, tourists are indifferent between buying a ticket and not going at all.

(c) Show that in the optimal solution, business travelers never buy their ticket prior to the flight and are just indifferent between doing this and buying when tourists buy.

(d) Describe fully the optimal price discrimination scheme under the assumption that they sell to both types. How does it depend on the underlying parameters  $\lambda$ ,  $\theta_B$ ,  $\theta_T$ , and  $c$ ?

(e) Under what circumstances will Air Shangri-la choose to serve only business travelers?

**14.C.9<sup>C</sup>** Consider a risk-averse individual who is an expected utility maximizer with a Bernoulli utility function over wealth  $u(\cdot)$ . The individual has initial wealth  $W$  and faces a probability  $\theta$  of suffering a loss of size  $L$ , where  $W > L > 0$ .

An insurance contract may be described by a pair  $(c_1, c_2)$ , where  $c_1$  is the amount of wealth the individual has in the event of no loss and  $c_2$  is the amount the individual has if a loss is suffered. That is, in the event no loss occurs the individual pays the insurance company an amount  $(W - c_1)$ , whereas if a loss occurs the individual receives a payment  $[c_2 - (W - L)]$  from the company.

(a) Suppose that the individual's only source of insurance is a risk-neutral monopolist (i.e., the monopolist seeks to maximize its expected profits). Characterize the contract the monopolist will offer the individual in the case in which the individual's probability of loss,  $\theta$ , is observable.

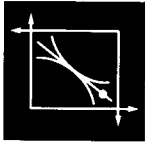
(b) Suppose, instead, that  $\theta$  is not observable by the insurance company (the individual knows  $\theta$ ). The parameter  $\theta$  can take one of two values  $\{\theta_L, \theta_H\}$ , where  $\theta_H > \theta_L > 0$  and  $\text{Prob}(\theta_L) = \lambda$ . Characterize the optimal contract offers of the monopolist. Can one speak of one type of insured individual being "rationed" in his purchases of insurance (i.e., he would want to purchase more insurance if allowed to at fair odds)? Intuitively, why does this rationing occur? [Hint: It might be helpful to draw a picture in  $(c_1, c_2)$ -space. To do so, start by locating the individual's endowment point, that is, what he gets if he does not purchase any insurance.]

(c) Compare your solution in (b) with your answer to Exercise 13.D.2.

**14.AA.1<sup>B</sup>** Show that  $[f_e(\pi|e)/f(\pi|e)]$  is increasing in  $\pi$  for all  $e \in [a, b] \subset \mathbb{R}$  if and only if for any  $e', e'' \in [a, b]$ , with  $e'' > e'$ ,  $[f(\pi|e'')/f(\pi|e')]$  is increasing in  $\pi$ .

**14.AA.2<sup>B</sup>** Consider a hidden action model with  $e \in [0, \bar{e}]$  and two outcomes  $\pi_H$  and  $\pi_L$ , with  $\pi_H > \pi_L$ . The probability of  $\pi_H$  given effort level  $e$  is  $f(\pi_H|e)$ . Give sufficient conditions for the first-order approach to be valid. Characterize the optimal contract when these conditions are satisfied.

**14.BB.1<sup>B</sup>** Try solving problem (14.BB.1) by first solving it while ignoring constraint (iv) and then arguing that the solution you derive to this "relaxed" problem is actually the solution to problem (14.BB.1).



# General Equilibrium

Part IV is devoted to an examination of competitive market economies from a *general equilibrium* perspective. Our use of the term “general equilibrium” refers both to a methodological point of view and to a substantive theory.

Methodologically, the general equilibrium approach has two central features. First, it views the economy as a *closed* and *interrelated* system in which we must simultaneously determine the equilibrium values of all variables of interest. Thus, when we evaluate the effects of a perturbation in the economic environment, the equilibrium levels of the entire set of endogenous variables in the economy needs to be recomputed. This stands in contrast to the *partial equilibrium* approach, where the impact on endogenous variables not directly related to the problem at hand is explicitly or implicitly disregarded.

A second central feature of the general equilibrium approach is that it aims at reducing the set of variables taken as exogenous to a small number of physical realities (e.g., the set of economic agents, the available technologies, the preferences and physical endowments of goods of various agents).

From a substantive viewpoint, general equilibrium theory has a more specific meaning: It is a theory of the determination of equilibrium prices and quantities in a system of perfectly competitive markets. This theory is often referred to as the *Walrasian theory* of markets [from L. Walras (1874)], and it is the object of our study in Part IV. The Walrasian theory of markets is very ambitious. It attempts no less than to predict the complete vector of final consumptions and productions using only the fundamentals of the economy (the list of commodities, the state of technology, preferences and endowments), the institutional assumption that a price is quoted for every commodity (including those that will not be traded at equilibrium), and the behavioral assumption of price taking by consumers and firms.

Strictly speaking, we introduced a particular case of the general equilibrium model in Chapter 10. There, we carried out an equilibrium and welfare analysis of perfectly competitive markets under the assumption that consumers had quasilinear preferences. In that setting, consumer demand functions do not display wealth effects (except for a single commodity, called the *numeraire*); as a consequence, the analysis of a single market (or small group of markets) could be pursued in a manner understandable as traditional partial equilibrium analysis. A good deal of what we do in Part IV

can be viewed as an attempt to extend the ideas of Chapter 10 to a world in which wealth effects are significant. The primary motivation for this is the increase in realism it brings. To make practical use of equilibrium analysis for studying the performance of an entire economy, or for evaluating policy interventions that affect large numbers of markets simultaneously, wealth effects, a primary source of linkages across markets, cannot be neglected, and therefore the general equilibrium approach is essential.

Although knowledge of the material discussed in Chapter 10 is not a strict prerequisite for Part IV, we nonetheless strongly recommend that you study it, especially Sections 10.B to 10.D. It constitutes an introduction to the main issues and provides a simple and analytically very useful example. We will see in the different chapters of Part IV that quite a number of the important results established in Chapter 10 for the quasilinear situation carry over to the case of general preferences. But many others do not. To understand why this may be so, recall from Chapters 4 and 10 that a group of consumers with quasilinear preferences (with respect to the same numeraire) admits the existence of a (normative) representative consumer. This is a powerful restriction on the behavior of aggregate demand that will not be available to us in the more general settings that we study here.

It is important to note that, relative to the analysis carried out in Part III, we incur a cost for accomplishing the task that general equilibrium sets itself to do: the assumptions of price-taking behavior and universal price quoting—that is, the existence of markets for every relevant commodity (with the implication of symmetric information)—are present in nearly all the theory studied in Part IV. Thus, in many respects, we are not going as deep as we did in Part III in the microanalysis of markets, of market failure, and of the strategic interdependence of market actors. The trade-off in conceptual structure between Parts III and IV reflects, in a sense, the current state of the frontier of microeconomic research.

The content of Part IV is organized into six chapters.

Chapter 15 presents a preliminary discussion. Its main purpose is to illustrate the issues that concern general equilibrium theory by means of three simple examples: the *two-consumer Edgeworth box economy*; the *one-consumer, one-firm economy*, and the *small open economy* model.

Chapters 16 and 17 constitute the heart of the formal analysis in Part IV. Chapter 16 presents the formal structure of the general equilibrium model and introduces two central concepts of the theory: the notions of *Pareto optimality* and *price-taking equilibrium* (and, in particular, *Walrasian equilibrium*). The chapter is devoted to the examination of the relationship between these two concepts. The emphasis is therefore *normative*, focusing on the welfare properties of price-taking equilibria. The core of the chapter is concerned with the formulation and proof of the two *fundamental theorems of welfare economics*.

In Chapter 17, the emphasis is, instead, on *positive* (or *descriptive*) properties of Walrasian equilibria. We study a number of questions pertaining to the predictive power of the Walrasian theory, including the existence, local and global uniqueness, and comparative statics behavior of Walrasian equilibria.

Chapters 18 to 20 explore extensions of the basic analysis presented in Chapters 16 and 17. Chapter 18 covers a number of topics whose origins lie in normative theory or the cooperative theory of games; these topics share the feature that they provide a deeper look at the foundations of price-taking equilibria by exploiting

properties derived from the mass nature of markets. We study the important *core equivalence theorem*, examine further the idea of Walrasian equilibria as the limit of noncooperative equilibria as markets grow large (a subject already broached in Section 12.F), and present two normative characterizations of Walrasian equilibria: one in terms of *envy-freeness* (or *anonymity*) and the other in terms of a *marginal productivity principle*. Appendix A of Chapter 18 offers a brief introduction to the cooperative theory of games.

Chapter 19 covers the modeling of uncertainty in a general equilibrium context. The ability to do this in a theoretically satisfying way has been one of the success stories of general equilibrium theory. The concepts of *contingent commodities*, *Arrow–Debreu equilibrium*, *sequential trade* (in a two-period setting), *Radner equilibrium*, *arbitrage*, *rational expectations equilibrium*, and *incomplete markets* are all introduced and studied here. The chapter provides a natural link to the modern theory of finance.

Chapter 20 considers the application of the general theory to dynamic competitive economies (but with no uncertainty) and also studies a number of issues specific to this environment. Notions such as *impatience*, *dynamic efficiency*, and *myopic* versus *overall utility maximization* are introduced. The chapter first analyzes dynamic representative consumer economies (including the *Ramsey–Solow model*), then generalizes to the case of a finite number of consumers, and concludes with a brief presentation of the *overlapping generations model*. In the process, we explore a wide range of dynamic behaviors. The chapter provides a natural link to macroeconomic theory.

The modern classics of general equilibrium theory are Debreu (1959) and Arrow and Hahn (1971). These texts provide further discussion of topics treated here. For extensions, we recommend the encyclopedic coverage of Arrow and Intriligator (1981, 1982, 1986) and Hildenbrand and Sonnenschein (1991). See also the more recent textbook account of Ellickson (1993). General equilibrium analysis has a very important applied dimension that we do not touch on in this book but that accounts in good part for the importance of the theory. For a review, we recommend Shoven and Whalley (1992).

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# General Equilibrium Theory:

## Some Examples

### 15.A Introduction

The purpose of this chapter is to present a preliminary discussion. In it, we describe and analyze three simple examples of general equilibrium models. These examples introduce some of the questions, concepts, and common techniques that will occupy us for the rest of Part IV.

In most economies, three basic economic activities occur: production, exchange, and consumption. In Section 15.B, we restrict our focus to exchange and consumption. We analyze the case of a *pure exchange economy*, in which no production is possible and the commodities that are ultimately consumed are those that individuals possess as *endowments*. Individuals trade these endowments among themselves in the marketplace for mutual advantage. The model we present is the simplest-possible exchange problem: two consumers trading two goods between each other. In this connection, we introduce an extremely handy graphical device, the *Edgeworth box*.

In Section 15.C, we introduce production by studying an economy formed by one firm and one consumer. Using this simple model, we explore how the production and consumption sides of the economy fit together.

In Section 15.D, we examine the production side of the economy in greater detail by discussing the allocation of resources among several firms. To analyze this issue in isolation, we study the case of a small open economy that takes the world prices of its outputs as fixed, a central model in international trade literature.

Section 15.E illustrates, by means of an example, some of the potential dangers of adopting a partial equilibrium perspective when a general equilibrium approach is called for.

As we noted in the introduction of Part IV, Chapter 10 contains another simple example of a general equilibrium model: that of an economy in which consumers have preferences admitting a quasilinear representation.

### 15.B Pure Exchange: The Edgeworth Box

A *pure exchange economy* (or, simply, an *exchange economy*) is an economy in which there are no production opportunities. The economic agents of such an economy are

consumers who possess initial stocks, or *endowments*, of commodities. Economic activity consists of trading and consumption.

The simplest economy with the possibility of profitable exchange is one with two commodities and two consumers. As it turns out, this case is amenable to analysis by a graphical device known as the *Edgeworth box*, which we use extensively in this section. Throughout, we assume that the two consumers act as price takers. Although this may not seem reasonable with only two traders, our aim here is to illustrate some of the features of general equilibrium models in the simplest-possible way.<sup>1</sup>

To begin, assume that there are two consumers, denoted by  $i = 1, 2$ , and two commodities, denoted by  $\ell = 1, 2$ . Consumer  $i$ 's consumption vector is  $x_i = (x_{1i}, x_{2i})$ ; that is, consumer  $i$ 's consumption of commodity  $\ell$  is  $x_{\ell i}$ . We assume that consumer  $i$ 's consumption set is  $\mathbb{R}_+^2$  and that he has a preference relation  $\succeq_i$  over consumption vectors in this set. Each consumer  $i$  is initially endowed with an amount  $\omega_{\ell i} \geq 0$  of good  $\ell$ . Thus, consumer  $i$ 's *endowment vector* is  $\omega_i = (\omega_{1i}, \omega_{2i})$ . The *total endowment* of good  $\ell$  in the economy is denoted by  $\bar{\omega}_\ell = \omega_{\ell 1} + \omega_{\ell 2}$ ; we assume that this quantity is strictly positive for both goods.

An *allocation*  $x \in \mathbb{R}_+^4$  in this economy is an assignment of a nonnegative consumption vector to each consumer:  $x = (x_1, x_2) = ((x_{11}, x_{21}), (x_{12}, x_{22}))$ . We say that an allocation is *feasible* for the economy if

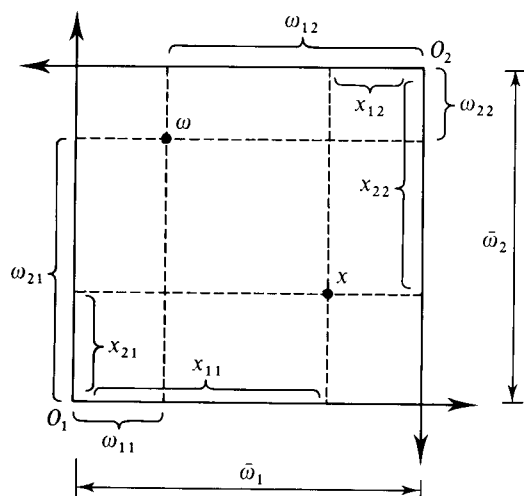
$$x_{\ell 1} + x_{\ell 2} \leq \bar{\omega}_\ell \quad \text{for } \ell = 1, 2, \quad (15.B.1)$$

that is, if the total consumption of each commodity is no more than the economy's aggregate endowment of it (note that in this notion of feasibility, we are implicitly assuming that there is free disposal of commodities).

The feasible allocations for which equality holds in (15.B.1) could be called *nonwasteful*. Nonwasteful feasible allocations can be depicted by means of an *Edgeworth box*, shown in Figure 15.B.1. In the Edgeworth box, consumer 1's quantities are measured in the usual way, with the southwest corner as the origin. In contrast, consumer 2's quantities are measured using the northeast corner as the origin. For both consumers, the vertical dimension measures quantities of good 2, and the horizontal dimension measures quantities of good 1. The length of the box is  $\bar{\omega}_1$ , the economy's total endowment of good 1; its height is  $\bar{\omega}_2$ , the economy's total endowment of good 2. Any point in the box represents a (nonwasteful) division of the economy's total endowment between consumers 1 and 2. For example, Figure 15.B.1 depicts the endowment vector  $\omega = ((\omega_{11}, \omega_{21}), (\omega_{12}, \omega_{22}))$  of the two consumers. Also depicted is another possible nonwasteful allocation,  $x = ((x_{11}, x_{21}), (x_{12}, x_{22}))$ ; the fact that it is nonwasteful means that  $(x_{12}, x_{22}) = (\bar{\omega}_1 - x_{11}, \bar{\omega}_2 - x_{21})$ .

As is characteristic of general equilibrium theory, the wealth of a consumer is not given exogenously. Rather, for any prices  $p = (p_1, p_2)$ , consumer  $i$ 's wealth equals the market value of his endowments of commodities,  $p \cdot \omega_i = p_1 \omega_{1i} + p_2 \omega_{2i}$ . Wealth levels are therefore determined by the values of prices. Hence, given the consumer's endowment vector  $\omega_i$ , his budget set can be viewed solely as a

1. Alternatively, we could assume that each consumer (perhaps better called a *consumer type*) stands, not for an individual, but for a large number of identical consumers. This would make the price-taking assumption more plausible; and with equal numbers of the two types of consumers, the analysis in this section would be otherwise unaffected.



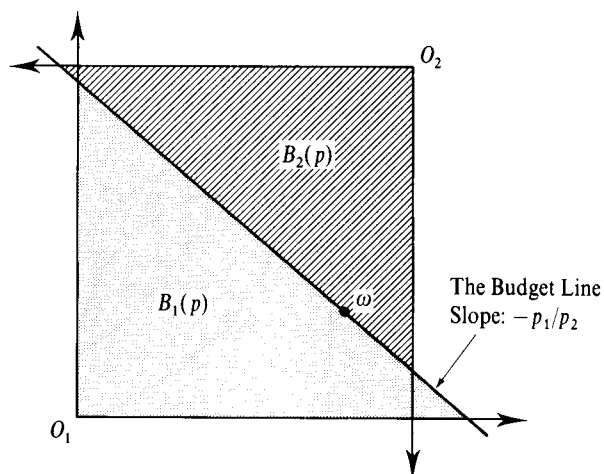
function of prices:

$$B_i(p) = \{x_i \in \mathbb{R}_+^2 : p \cdot x_i \leq p \cdot \omega_i\}.$$

The budget sets of the two consumers can be represented in the Edgeworth box in a simple manner. To do so, we draw a line, known as the *budget line*, through the endowment point  $\omega$  with slope  $-(p_1/p_2)$ , as shown in Figure 15.B.2. Consumer 1's budget set consists of all the nonnegative vectors below and to the left of this line (the shaded set). Consumer 2's budget set, on the other hand, consists of all the vectors above and to the right of this same line which give consumer 2 nonnegative consumption levels (the hatched set).<sup>2</sup> Observe that only allocations on the budget line are affordable to both consumers simultaneously at prices  $(p_1, p_2)$ .<sup>3</sup>

We can also depict the preferences  $\succeq_i$  of each consumer  $i$  in the Edgeworth box, as in Figure 15.B.3. Except where otherwise noted, we assume that  $\succeq_i$  is strictly convex, continuous, and strongly monotone (see Sections 3.B and 3.C for discussion of these conditions).

Figure 15.B.4 illustrates how the consumption vector demanded by consumer 1 can be determined for any price vector  $p$ . Given  $p$ , the consumer demands his most preferred point in  $B_1(p)$ , which can be expressed using his demand function as  $x_1(p, p \cdot \omega_1)$  (this is the same demand function studied in Chapters 2 to 4; here wealth is  $w_1 = p \cdot \omega_1$ ). In Figure 15.B.5, we see that as the price vector  $p$  varies, the budget line pivots around the endowment point  $\omega$ , and the demanded consumptions trace out a curve, denoted by  $OC_1$ , that is called the *offer curve* of consumer 1. Note that this curve passes through the endowment point. Because at every  $p$  the endowment vector  $\omega_1 = (\omega_{11}, \omega_{21})$  is affordable to consumer 1, it follows that this consumer must find every point on his offer curve at least as good as his endowment point.

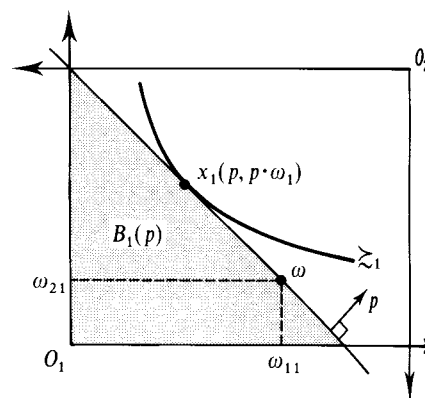
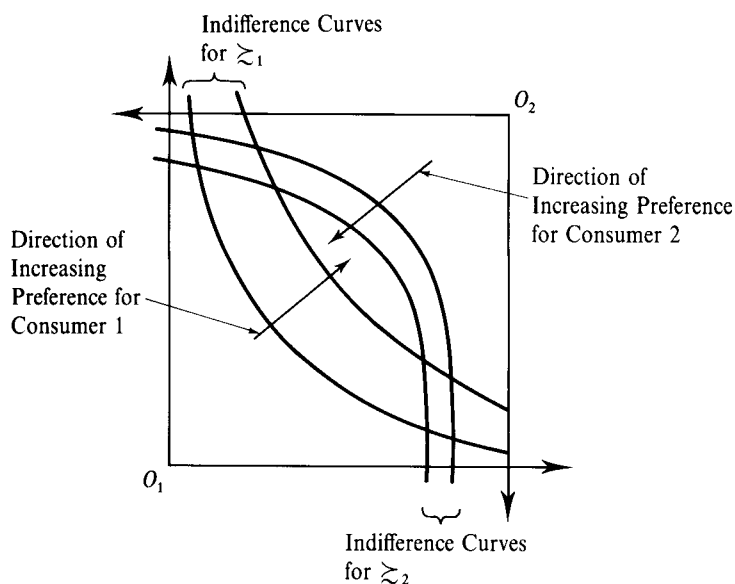


**Figure 15.B.1 (left)**  
An Edgeworth box.

**Figure 15.B.2 (right)**  
Budget sets.

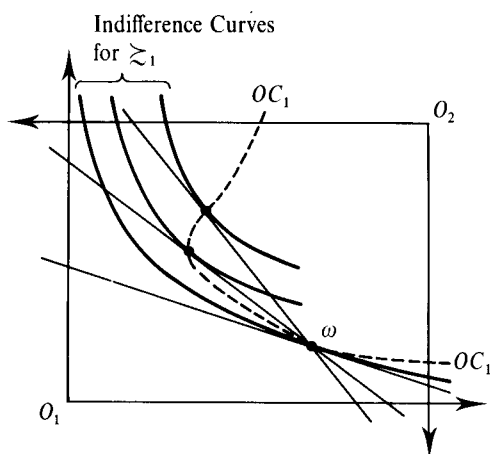
2. Note, in particular, that the budget sets of the consumers may well extend outside the box.

3. There are other feasible allocations that are simultaneously affordable; but in these allocations some resources are not consumed by either consumer, and thus they cannot be depicted in an Edgeworth box. Because of the nonsatiation assumption to be made on preferences, we will not have to worry about such allocations.



**Figure 15.B.3 (top left)**  
Preferences in the Edgeworth box.

**Figure 15.B.4 (top right)**  
Optimal consumption for consumer 1 at prices  $p$ .

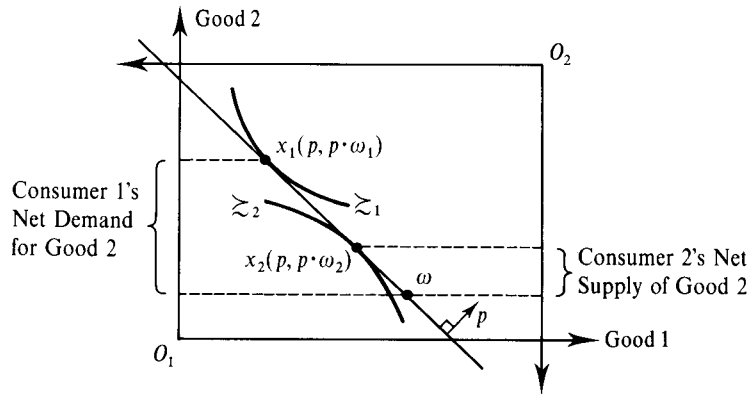


**Figure 15.B.5 (bottom)**  
Consumer 1's offer curve.

This implies that the consumer's offer curve lies within the upper contour set of  $\omega_1$  and that, if indifference curves are smooth, the offer curve must be tangent to the consumer's indifference curve at the endowment point.

Figure 15.B.6 represents the demanded bundles of the two consumers at some arbitrary price vector  $p$ . Note that the demands expressed by the two consumers are not compatible. The total demand for good 2 exceeds its total supply in the economy  $\bar{\omega}_2$ , whereas the total demand for good 1 is strictly less than its endowment  $\bar{\omega}_1$ . Put somewhat differently, consumer 1 is a *net demander* of good 2 in the sense that he wants to consume more than his endowment of that commodity. Although consumer 2 is willing to be a *net supplier* of that good (he wants to consume less than his endowment), he is not willing to supply enough to satisfy consumer 1's needs. Good 2 is therefore in *excess demand* in the situation depicted in the figure. In contrast, good 1 is in *excess supply*.

At a market equilibrium where consumers take prices as given, markets should clear. That is, the consumers should be able to fulfill their desired purchases and

**Figure 15.B.6**

A price vector with excess demand for good 2 and excess supply for good 1.

sales of commodities at the going market prices. Thus, if one consumer wishes to be a *net demander* of some good, the other must be a *net supplier* of this good in exactly the same amount; that is, demand should equal supply. This gives us the notion of equilibrium presented in Definition 15.B.1.

**Definition 15.B.1:** A *Walrasian* (or *competitive*) *equilibrium* for an Edgeworth box economy is a price vector  $p^*$  and an allocation  $x^* = (x_1^*, x_2^*)$  in the Edgeworth box such that for  $i = 1, 2$ ,

$$x_i^* \succeq_i x'_i \quad \text{for all } x'_i \in B_i(p^*).$$

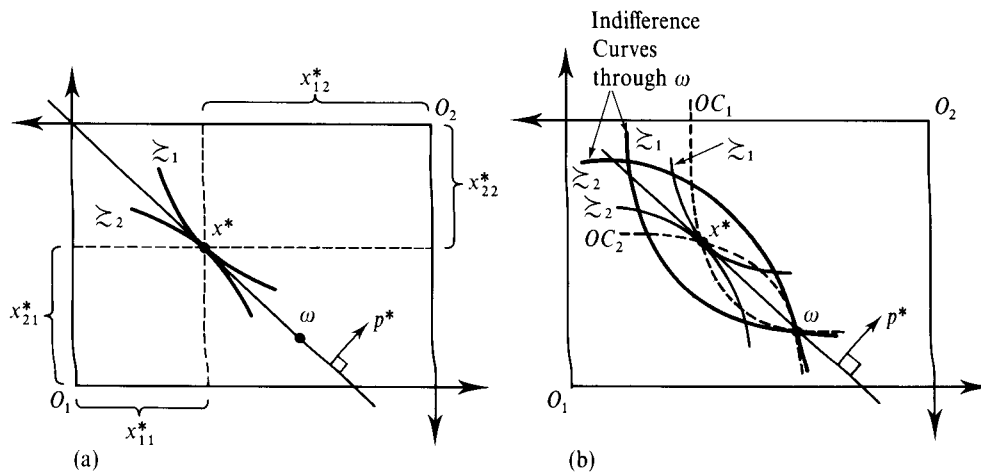
A Walrasian equilibrium is depicted in Figure 15.B.7. In Figure 15.B.7(a), we represent the equilibrium price vector  $p^*$  and the equilibrium allocation  $x^* = (x_1^*, x_2^*)$ . Each consumer  $i$ 's demanded bundle at price vector  $p^*$  is  $x_i^*$ , and one consumer's net demand for a good is exactly matched by the other's net supply. Figure 15.B.7(b) adds to the depiction the consumers' offer curves and their indifference curves through  $\omega$ . Note that at any equilibrium, the offer curves of the two consumers intersect. In fact, *any* intersection of the consumers' offer curves at an allocation different from the endowment point  $\omega$  corresponds to an equilibrium because if  $x^* = (x_1^*, x_2^*)$  is any such point of intersection, then  $x_i^*$  is the optimal consumption bundle for each consumer  $i$  for the budget line that goes through the two points  $\omega$  and  $x^*$ .

In Figure 15.B.8, we show a Walrasian equilibrium where the equilibrium allocation lies on the boundary of the Edgeworth box. Once again, at price vector  $p^*$ , the two consumers' demands are compatible.

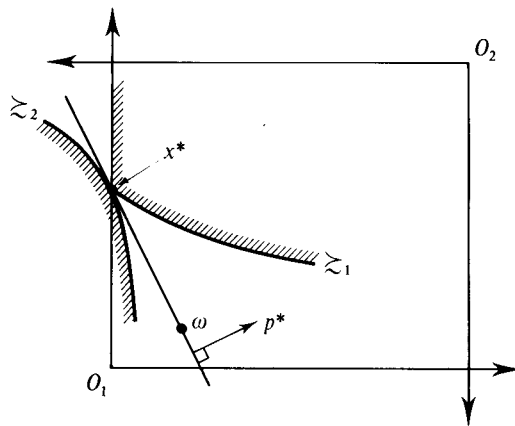
Note that each consumer's demand is homogeneous of degree zero in the price vector  $p = (p_1, p_2)$ ; that is, if prices double, then the consumer's wealth also doubles and his budget set remains unchanged. Thus, from Definition 15.B.1, we see that if  $p^* = (p_1^*, p_2^*)$  is a Walrasian equilibrium price vector, then so is  $\alpha p^* = (\alpha p_1^*, \alpha p_2^*)$  for any  $\alpha > 0$ . In short, only the *relative* prices  $p_1^*/p_2^*$  are determined in an equilibrium.

**Example 15.B.1:** Suppose that each consumer  $i$  has the Cobb–Douglas utility function  $u_i(x_{1i}, x_{2i}) = x_{1i}^\alpha x_{2i}^{1-\alpha}$ . In addition, endowments are  $\omega_1 = (1, 2)$  and  $\omega_2 = (2, 1)$ . At prices  $p = (p_1, p_2)$ , consumer 1's wealth is  $(p_1 + 2p_2)$  and therefore his demands lie on the offer curve (recall the derivation in Example 3.D.1):

$$OC_1(p) = \left( \frac{\alpha(p_1 + 2p_2)}{p_1}, \frac{(1 - \alpha)(p_1 + 2p_2)}{p_2} \right).$$

**Figure 15.B.7 (top)**

(a) A Walrasian equilibrium.  
(b) The consumer's offer curves intersect at the Walrasian equilibrium allocation

**Figure 15.B.8 (bottom)**

A Walrasian equilibrium allocation on the boundary of the Edgeworth box.

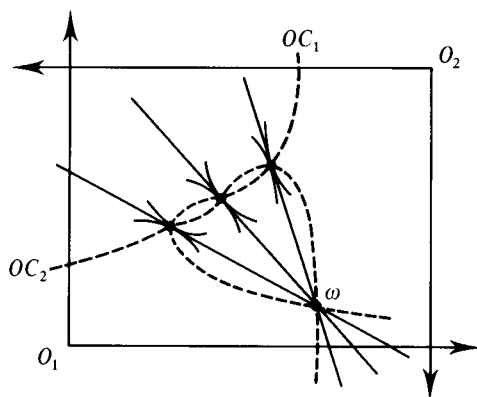
Observe that the demands for the first and the second good are, respectively, decreasing and increasing with  $p_1$ . This is how we have drawn  $OC_1$  in Figure 15.B.7(b). Similarly,  $OC_2(p) = (\alpha(2p_1 + p_2)/p_1, (1 - \alpha)(2p_1 + p_2)/p_2)$ . To determine the Walrasian equilibrium prices, note that at these prices the total amount of good 1 consumed by the two consumers must equal 3 ( $= \omega_{11} + \omega_{12}$ ). Thus,

$$\frac{\alpha(p_1^* + 2p_2^*)}{p_1^*} + \frac{\alpha(2p_1^* + p_2^*)}{p_1^*} = 3.$$

Solving this equation yields

$$\frac{p_1^*}{p_2^*} = \frac{\alpha}{1 - \alpha}. \quad (15.B.2)$$

Observe that at any prices  $(p_1^*, p_2^*)$  satisfying condition (15.B.2), the market for good 2 clears as well (you should verify this). This is a general feature of an Edgeworth box economy: To determine equilibrium prices we need only determine prices at which one of the markets clears; the other market will necessarily clear at these prices. This point can be seen graphically in the Edgeworth box: Because both consumers' demanded bundles lie on the same budget line, if the amounts of commodity 1 demanded are compatible, then so must be those for commodity 2. (See also Exercise 15.B.1.) ■



**Figure 15.B.9**  
Multiple Walrasian equilibria.

The Edgeworth box, simple as it is, is remarkably powerful. There are virtually no phenomena or properties of general equilibrium exchange economies that cannot be depicted in it. Consider, for example, the issue of the uniqueness of Walrasian equilibrium. In Chapter 10, we saw that if there is a numeraire commodity relative to which preferences admit a quasilinear representation, then (with strict convexity of preferences) the equilibrium consumption allocation and relative prices are unique. In Figure 15.B.7, we also have uniqueness (see Exercise 15.B.2 for a more explicit discussion). Yet, as the Edgeworth box in Figure 15.B.9 shows, this property does not generalize. In that figure, preferences (which are entirely nonpathological) are such that the offer curves change curvature and interlace several times. In particular, they intersect for prices such that  $p_1/p_2$  is equal to  $\frac{1}{2}$ , 1, and 2. For the sake of completeness, we present an analytical example with the features of the figure.

**Example 15.B.2:** Let the two consumers have utility functions

$$u_1(x_{11}, x_{21}) = x_{11} - \frac{1}{8}x_{21}^{-8} \text{ and } u_2(x_{12}, x_{22}) = -\frac{1}{8}x_{12}^{-8} + x_{22}.$$

Note that the utility functions are quasilinear (which, in particular, facilitates the computation of demand), but with respect to *different* numeraire. The endowments are  $\omega_1 = (2, r)$  and  $\omega_2 = (r, 2)$ , where  $r$  is chosen to guarantee that the equilibrium prices turn out to be round numbers. Precisely,  $r = 2^{8/9} - 2^{1/9} > 0$ . In Exercise 15.B.5, you are asked to compute the offer curves of the two consumers. They are:

$$OC_1(p_1, p_2) = \left( 2 + r \left( \frac{p_2}{p_1} \right) - \left( \frac{p_2}{p_1} \right)^{8/9}, \left( \frac{p_2}{p_1} \right)^{-1/9} \right) \gg 0$$

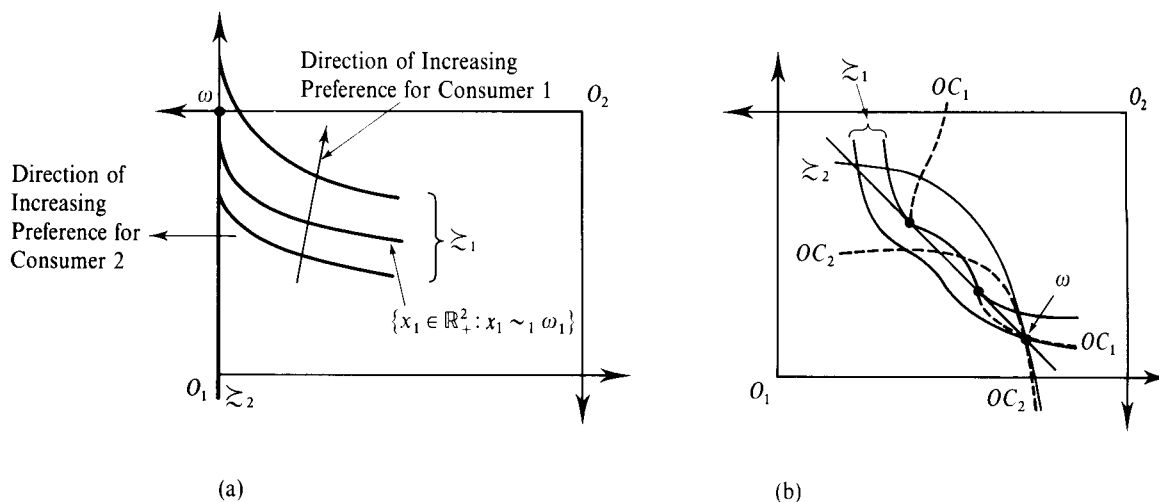
and

$$OC_2(p_1, p_2) = \left( \left( \frac{p_1}{p_2} \right)^{-1/9}, 2 + r \left( \frac{p_1}{p_2} \right) - \left( \frac{p_1}{p_2} \right)^{8/9} \right) \gg 0.$$

Note that, as illustrated in Figure 15.B.9, and in contrast with Example 15.B.1, consumer 1's demand for good 1 (and symmetrically for consumer 2) may be increasing in  $p_1$ .

To compute the equilibria it is sufficient to solve the equation that equates the total demand of the second good to its total supply, or

$$\left( \frac{p_2}{p_1} \right)^{-1/9} + 2 + r \left( \frac{p_1}{p_2} \right) - \left( \frac{p_1}{p_2} \right)^{8/9} = 2 + r.$$



**Figure 15.B.10** (a) and (b): Two examples of nonexistence of Walrasian equilibrium.

Recalling the value of  $r$ , this equation has three solutions for  $p_1/p_2$ : 2, 1, and  $\frac{1}{2}$  (you should check this). ■

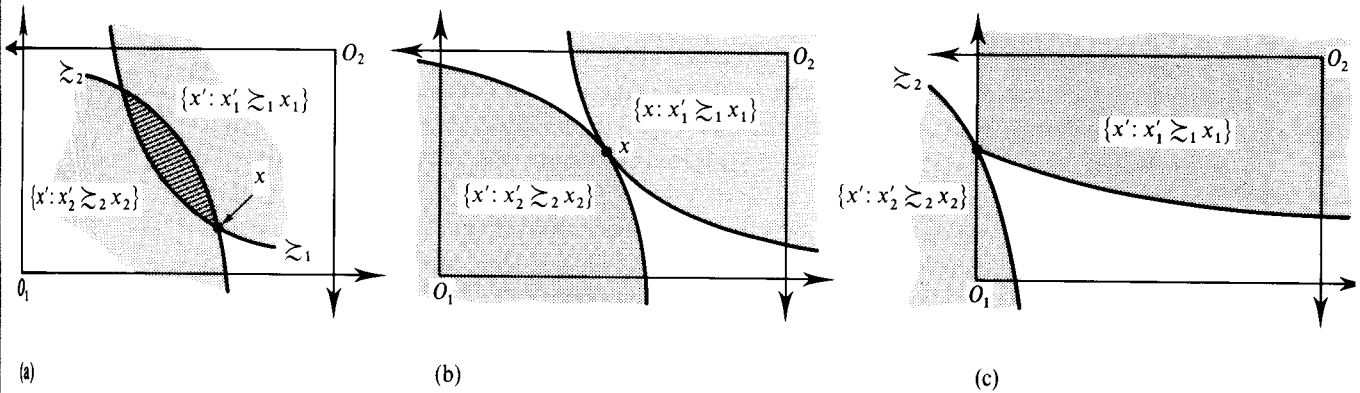
It may also happen that a pure exchange economy does not have *any* Walrasian equilibria. For example, Figure 15.B.10(a) depicts a situation in which the endowment lies on the boundary of the Edgeworth box (in the northwest corner). Consumer 2 has all the endowment of good 1 and desires only good 1. Consumer 1 has all the endowment of good 2 and his indifference set containing  $\omega_1$ ,  $\{x_1 \in \mathbb{R}_+^2 : x_1 \sim_1 \omega_1\}$ , has an infinite slope at  $\omega_1$  (note, however, that at  $\omega_1$ , consumer 1 would strictly prefer receiving more of good 1). In this situation, there is no price vector  $p^*$  at which the consumers' demands are compatible. If  $p_2/p_1 > 0$  then consumer 2 optimal demand is to keep his initial bundle  $\omega_2$ , whereas the initial bundle  $\omega_1$  is never consumer 1's optimal demand (no matter how large the relative price of the first good, consumer 1 always wishes to buy a strictly positive amount of it). On the other hand, consumer 1's demand for good 2 is infinite when  $p_2/p_1 = 0$ . Note for future reference that consumer 2's preferences in this example are not strongly monotone.

Figure 15.B.10(b) depicts a second example of nonexistence. There, consumer 1's preferences are nonconvex. As a result, consumer 1's offer curve is disconnected, and there is no crossing point of the two consumers' offer curves (other than the endowment point, which is not an equilibrium allocation here).

In Chapter 17, we will study the conditions under which the existence of a Walrasian equilibrium is assured.

### *Welfare Properties of Walrasian Equilibria*

A central question in economic theory concerns the welfare properties of equilibria. Here we shall focus on the notion of Pareto optimality, which we have already encountered in Chapter 10 (see, in particular, Section 10.B). An economic outcome is *Pareto optimal* (or *Pareto efficient*) if there is no alternative feasible outcome at which every individual in the economy is at least as well off and some individual is



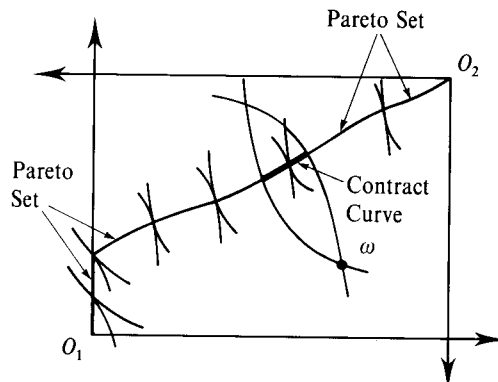
**Figure 15.B.11** (a) Allocation  $x$  is not Pareto optimal. (b) Allocation  $x$  is Pareto optimal. (c) Allocation  $x$  is Pareto optimal.

strictly better off. Definition 15.B.2 expresses this idea in the setting of our two-consumer, pure exchange economy.

**Definition 15.B.2:** An allocation  $x$  in the Edgeworth box is *Pareto optimal* (or *Pareto efficient*) if there is no other allocation  $x'$  in the Edgeworth box with  $x'_i \succsim_i x_i$  for  $i = 1, 2$  and  $x'_i \succ_i x_i$  for some  $i$ .

Figure 15.B.11(a) depicts an allocation  $x$  that is not Pareto optimal. Any allocation in the interior of the crosshatched region of the figure, the intersection of the sets  $\{x'_1 \in \mathbb{R}_+^2 : x'_1 \succsim_1 x_1\}$  and  $\{x'_2 \in \mathbb{R}_+^2 : x'_2 \succsim_2 x_2\}$  within the Edgeworth box, is a feasible allocation that makes both consumers strictly better off than at  $x$ . The allocation  $x$  depicted in Figure 15.B.11(b), on the other hand, is Pareto optimal because the intersection of the sets  $\{x'_i \in \mathbb{R}_+^2 : x'_i \succsim_i x_i\}$  for  $i = 1, 2$  consists only of the point  $x$ . Note that if a Pareto optimal allocation  $x$  is an interior point of the Edgeworth box, then the consumers' indifference curves through  $x$  must be tangent (assuming that they are smooth). Figure 15.B.11(c) depicts a Pareto optimal allocation  $x$  that is not interior; at such a point, tangency need not hold.

The set of all Pareto optimal allocations is known as the *Pareto set*. An example is illustrated in Figure 15.B.12. The figure also displays the *contract curve*, the part of the



**Figure 15.B.12**  
The Pareto set and the contract curve.

Pareto set where both consumers do at least as well as at their initial endowments. The reason for this term is that we might expect any bargaining between the two consumers to result in an agreement to trade to some point on the contract curve; these are the only points at which both of them do as well as at their initial endowments and for which there is no alternative trade that can make both consumers better off.

We can now verify a simple but important fact: *Any Walrasian equilibrium allocation  $x^*$  necessarily belongs to the Pareto set.* To see this, note that by the definition of a Walrasian equilibrium the budget line separates the two at-least-as-good-as sets associated with the equilibrium allocation, as seen in Figures 15.B.7(a) and 15.B.8. The only point in common between these two sets is  $x^*$  itself. Thus, at any competitive allocation  $x^*$ , there is no alternative feasible allocation that can benefit one consumer without hurting the other. The conclusion that Walrasian allocations yield Pareto optimal allocations is an expression of the *first fundamental theorem of welfare economics*, a result that, as we shall see in Chapter 16, holds with great generality. Note, moreover, that since each consumer must be at least as well off in a Walrasian equilibrium as by simply consuming his endowment, any Walrasian equilibrium lies in the contract curve portion of the Pareto set.

The first fundamental welfare theorem provides, for competitive market economies, a formal expression of Adam Smith's "invisible hand." Under perfectly competitive conditions, any equilibrium allocation is a Pareto optimum, and the only possible welfare justification for intervention in the economy is the fulfillment of distributional objectives.

The *second fundamental theorem of welfare economics*, which we also discuss extensively in Chapter 16, offers a (partial) converse result. Roughly put, it says that *under convexity assumptions (not required for the first welfare theorem), a planner can achieve any desired Pareto optimal allocation by appropriately redistributing wealth in a lump-sum fashion and then "letting the market work."* Thus, the second welfare theorem provides a theoretical affirmation for the use of competitive markets in pursuing distributional objectives.

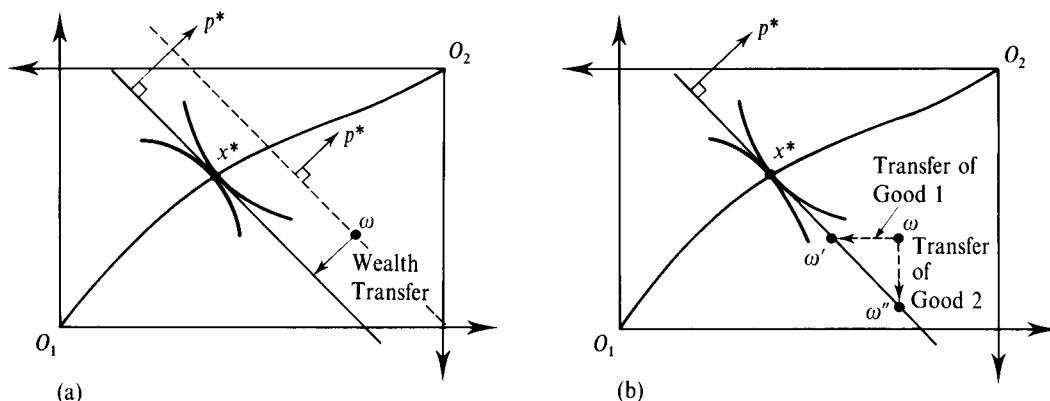
Definition 15.B.3 is a more formal statement of the concept of an equilibrium with lump-sum wealth redistribution.

**Definition 15.B.3:** An allocation  $x^*$  in the Edgeworth box is supportable as an *equilibrium with transfers* if there is a price system  $p^*$  and wealth transfers  $T_1$  and  $T_2$  satisfying  $T_1 + T_2 = 0$ , such that for each consumer  $i$  we have

$$x_i^* \succeq_i x'_i \quad \text{for all } x'_i \in \mathbb{R}_+^2 \quad \text{such that} \quad p^* \cdot x'_i \leq p^* \cdot \omega_i + T_i.$$

Note that the transfers sum to zero in Definition 15.B.3; the planner runs a balanced budget, merely redistributing wealth between the consumers.

Equipped with Definition 15.B.3, we can state more formally a version of the second welfare theorem as follows: if the preferences of the two consumers in the Edgeworth box are continuous, convex, and strongly monotone, then *any Pareto optimal allocation is supportable as an equilibrium with transfers.* This result is illustrated in Figure 15.B.13(a), where the consumer's endowments are at point  $\omega$ . Suppose that for distributional reasons, the socially desired allocation is the Pareto optimal allocation  $x^*$ . Then if a tax authority constructs a transfer of wealth between the two consumers that shifts the budget line to the location



**Figure 15.B.13** The second fundamental welfare theorem. (a) Using wealth transfers. (b) Using transfers of endowments.

indicated in the figure, the price vector  $p^*$  clears the markets for the two goods, and allocation  $x^*$  results.

Note that this wealth transfer may also be accomplished by directly transferring endowments. As Figure 15.B.13(b) illustrates, a transfer of good 1 that moves the endowment vector to  $\omega'$  will have the price vector  $p^*$  and allocation  $x^*$  as a Walrasian equilibrium. A transfer of good 2 that changes endowments to  $\omega''$  does so as well. In fact, if all commodities can be easily transferred, then we could equally well move the endowment vector directly to allocation  $x^*$ . From this new endowment point, the Walrasian equilibrium involves no trade.<sup>4</sup>

Figure 15.B.14 shows that the second welfare theorem may fail to hold when preferences are not convex. In the figure,  $x^* = (x_1^*, x_2^*)$  is a Pareto optimal allocation that is not supportable as an equilibrium with transfers. At the budget line with the property that consumer 2 would demand  $x_2^*$ , consumer 1 would prefer a point other than  $x_1^*$  (such as  $x_1'$ ). Convexity, as it turns out, is a critical assumption for the second welfare theorem.

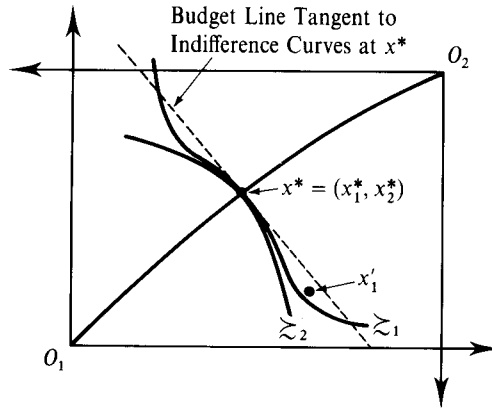
A failure of the second welfare theorem of a different kind is illustrated in Figure 15.B.10(a). There, the initial endowment allocation  $\omega$  is a Pareto optimal allocation, but it cannot be supported as an equilibrium with transfers (you should check this). In this case, it is the assumption that consumers' preferences are strongly monotone that is violated.

For further illustrations of Edgeworth box economies see, for example, Newman (1965).

## 15.C The One-Consumer, One-Producer Economy

We now introduce the possibility of production. To do so in the simplest-possible setting, we suppose that there are two price-taking economic agents, a single

4. In practice, endowments may be difficult to transfer (e.g., human capital), and so the ability to use wealth transfers (or transfers of only a limited number of commodities) may be important. It is worth observing that there is one attractive feature of transferring endowments directly to the desired Pareto optimal allocation: we can be assured that  $x^*$  is the *unique* Walrasian equilibrium allocation after the transfers (strictly speaking this requires a strict convexity assumption on preferences).



**Figure 15.B.14**  
Failure of the second welfare theorem with nonconvex preferences.

consumer and a single firm, and two goods, the labor (or leisure) of the consumer and a consumption good produced by the firm.<sup>5</sup>

The consumer has continuous, convex, and strongly monotone preferences  $\succsim$  defined over his consumption of leisure  $x_1$  and the consumption good  $x_2$ . He has an endowment of  $\bar{L}$  units of leisure (e.g., 24 hours in a day) and no endowment of the consumption good.

The firm uses labor to produce the consumption good according to the increasing and strictly concave production function  $f(z)$ , where  $z$  is the firm's labor input. Thus, to produce output, the firm must hire the consumer, effectively purchasing some of the consumer's leisure from him. We assume that the firm seeks to maximize its profits taking market prices as given. Letting  $p$  be the price of its output and  $w$  be the price of labor, the firm solves

$$\text{Max}_{z \geq 0} \quad pf(z) - wz. \quad (15.C.1)$$

Given prices  $(p, w)$ , the firm's optimal labor demand is  $z(p, w)$ , its output is  $q(p, w)$ , and its profits are  $\pi(p, w)$ .

As we noted in Chapter 5, firms are owned by consumers. Thus, we assume that the consumer is the sole owner of the firm and receives the profits earned by the firm  $\pi(p, w)$ . As with the price-taking assumption, the idea of the consumer being hired by his own firm through an anonymous labor market may appear strange in this model with only two agents. Nevertheless, bear with us; our aim is to illustrate the workings of more complicated many-consumer general equilibrium models in the simplest-possible way.<sup>6</sup>

Letting  $u(x_1, x_2)$  be a utility function representing  $\succsim$ , the consumer's problem given prices  $(p, w)$  is

$$\begin{aligned} \text{Max}_{(x_1, x_2) \in \mathbb{R}_+^2} \quad & u(x_1, x_2) \\ \text{s.t.} \quad & px_2 \leq w(\bar{L} - x_1) + \pi(p, w). \end{aligned} \quad (15.C.2)$$

5. One-consumer economies are sometimes referred to as *Robinson Crusoe economies*.

6. The point made in footnote 1 can be repeated here: we could imagine that the firm and the consumer stand for a large number of identical firms and consumers. We comment a bit more on this interpretation at the end of this Section.

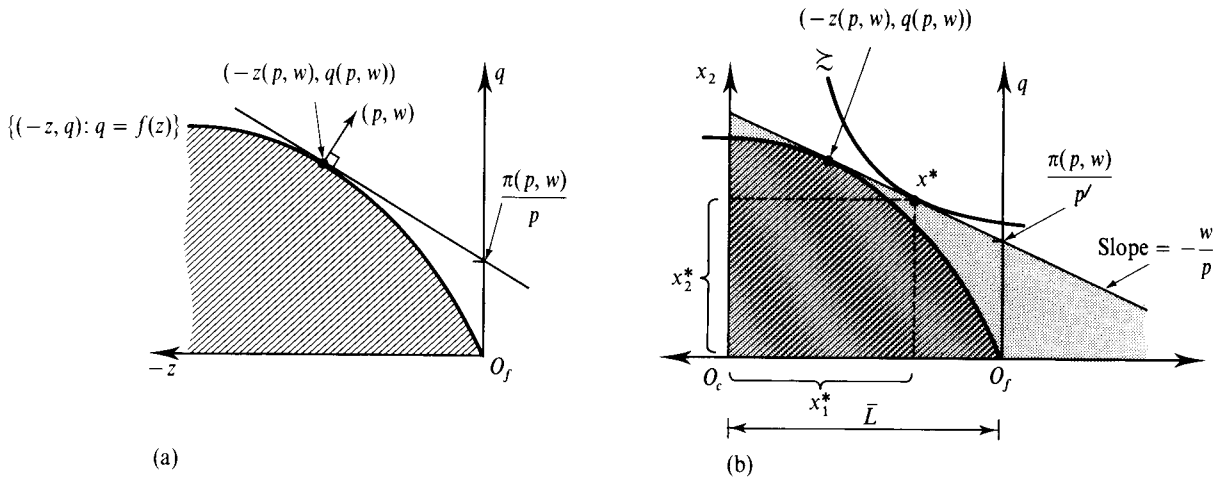


Figure 15.C.1 (a) The firm's problem. (b) The consumer's problem.

The budget constraint in (15.C.2) reflects the two sources of the consumer's purchasing power: If the consumer supplies an amount  $(\bar{L} - x_1)$  of labor when prices are  $(p, w)$ , then the total amount he can spend on the consumption good is his labor earnings  $w(\bar{L} - x_1)$  plus the profit distribution from the firm  $\pi(p, w)$ . The consumer's optimal demands in problem (15.C.2) for prices  $(p, w)$  are denoted by  $(x_1(p, w), x_2(p, w))$ .

A Walrasian equilibrium in this economy involves a price vector  $(p^*, w^*)$  at which the consumption and labor markets clear; that is, at which

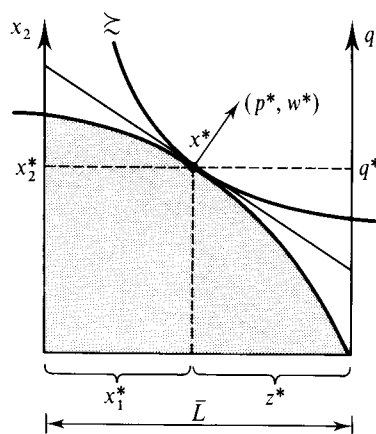
$$x_2(p^*, w^*) = q(p^*, w^*) \quad (15.C.3)$$

and

$$z(p^*, w^*) = \bar{L} - x_1(p^*, w^*) \quad (15.C.4)$$

Figure 15.C.1 illustrates the working of this one-consumer, one-firm economy. Figure 15.C.1(a) depicts the firm's problem. As in Chapter 5, we measure the firm's use of labor input on the horizontal axis as a negative quantity. Its output is depicted on the vertical axis. The production set associated with the production function  $f(z)$  is also shown, as are the profit-maximizing input and output levels at prices  $(p, w)$ ,  $z(p, w)$  and  $q(p, w)$ , respectively.

Figure 15.C.1(b) adapts this diagram to represent the consumer's problem. Leisure and consumption levels are measured from the origin denoted  $O_c$  at the lower-left-hand corner of the diagram, which is determined by letting the length of the segment  $[O_c, O_f]$  be equal to  $\bar{L}$ , the total labor endowment. The figure depicts the consumer's (shaded) budget set given prices  $(p, w)$  and profits  $\pi(p, w)$ . Note that if the consumer consumes  $\bar{L}$  units of leisure then since he sells no labor, he can purchase  $\pi(p, w)/p$  units of the consumption good. Thus, the budget line must cut the vertical  $q$ -axis at height  $\pi(p, w)/p$ . In addition, for each unit of labor he sells, the consumer earns  $w$  and can therefore afford to purchase  $w/p$  units of  $x_2$ . Hence, the budget line has slope  $-(w/p)$ . Observe that the consumer's budget line is exactly the isoprofit line associated with the solution to the firm's profit-maximization problem in Figure 15.C.1(a), that is, the set of points  $\{(-z, q): pq - wz = \pi(p, w)\}$  that yield profits of  $\pi(p, w)$ .



**Figure 15.C.2**  
A Walrasian equilibrium.

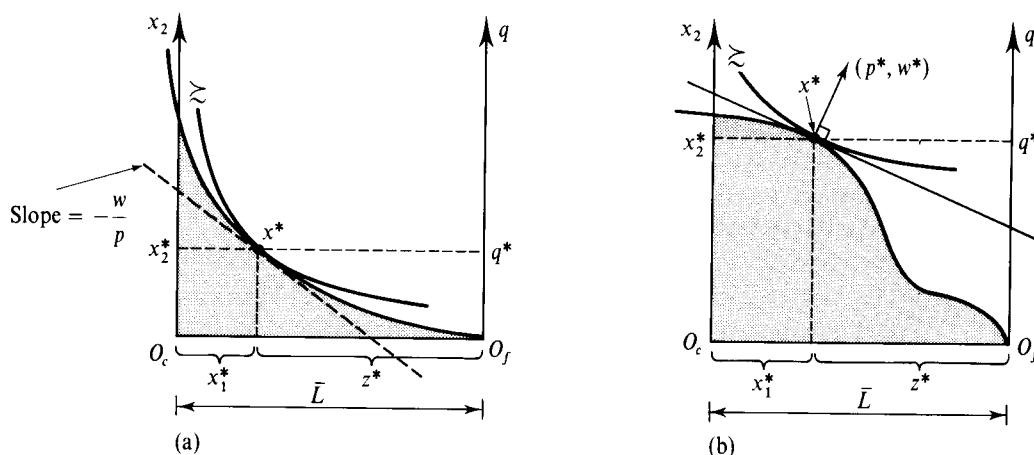
The prices depicted in Figure 15.C.1(b) are not equilibrium prices; at these prices, there is an excess demand for labor (the firm wants more labor than the consumer is willing to supply) and an excess supply of the produced good. An equilibrium price vector  $(p^*, w^*)$  that clears the markets for the two goods is depicted in Figure 15.C.2.

There is a very important fact to notice from Figure 15.C.2: *A particular consumption–leisure combination can arise in a competitive equilibrium if and only if it maximizes the consumer’s utility subject to the economy’s technological and endowment constraints.* That is, the Walrasian equilibrium allocation is the same allocation that would be obtained if a planner ran the economy in a manner that maximized the consumer’s well-being. Thus, we see here an expression of the fundamental theorems of welfare economics: Any Walrasian equilibrium is Pareto optimal, and the Pareto optimal allocation is supportable as a Walrasian equilibrium.<sup>7</sup>

The indispensability of convexity for the second welfare theorem can again be observed in Figure 15.C.3(a). There, the allocation  $x^*$  maximizes the welfare of the consumer, but for the only value of relative prices that could support  $x^*$  as a utility-maximizing bundle, the firm does not maximize profits even locally (i.e., at the relative prices  $w/p$ , there are productions arbitrarily close to  $x^*$  yielding higher profits). In contrast, the first welfare theorem remains applicable even in the presence of nonconvexities. As Figure 15.C.3(b) suggests, any Walrasian equilibrium maximizes the well-being of the consumer in the feasible production set.

Under certain circumstances, the model studied in this section can be rigorously justified as representing the outcome of a more general economy by interpreting the “firm” as a representative producer (see Section 5.E) and the “consumer” as a representative consumer (see Section 4.D). The former is always possible, but the latter—that is, the existence of a (normative) representative consumer—requires strong conditions. If, however, the economy

7. In a single-consumer economy, the test for Pareto optimality reduces to the question of whether the well-being of the single consumer is being maximized (subject to feasibility constraints). Note that given the convexity of preferences and the strict convexity of the aggregate production set assumed here, there is a unique Pareto optimal consumption vector (and therefore a unique equilibrium).



**Figure 15.C.3** (a) Failure of the second welfare theorem with a nonconvex technology. (b) The first welfare theorem applies even with a nonconvex technology.

is composed of many consumers with identical concave utility functions and identical initial endowments, and if society has a strictly concave social welfare function in which these consumers are treated symmetrically, then a (normative) representative consumer exists who has the same utility function as the consumers over levels of per capita consumption.<sup>8</sup> (We can also think of the representative firm's input and output choices as being on a per capita basis). For more general conditions under which a representative consumer exists, see Section 4.D.

## 15.D The $2 \times 2$ Production Model

In this section, we discuss an example that concentrates on general equilibrium effects in production.

To begin, consider an economy in which the production sector consists of  $J$  firms. Each firm  $j$  produces a consumer good  $q_j$  directly from a vector of  $L$  primary (i.e., nonproduced) inputs, or *factors*,  $z_j = (z_{1j}, \dots, z_{Lj}) \geq 0$ .<sup>9</sup> Firm  $j$ 's production takes place by means of a concave, strictly increasing, and differentiable production function  $f_j(z_j)$ . Note that there are no intermediate goods (i.e., produced goods that are used as inputs). The economy has total endowments of the  $L$  factor inputs,  $(\bar{z}_1, \dots, \bar{z}_L) \gg 0$ . These endowments are initially owned by consumers and have a use only as production inputs (i.e., consumers do not wish to consume them).

To concentrate on the factor markets of the economy, we suppose that the prices of the  $J$  produced consumption goods are fixed at  $p = (p_1, \dots, p_J)$ . The leading example for this assumption is that of a small open economy whose trading decisions in the world markets for consumption goods have little effect on the world prices of

8. To see this, note that an equal distribution of wealth (which is what occurs here in the absence of any wealth transfers given the identical endowments of the consumers) maximizes social welfare for any price vector and aggregate wealth level.

9. Some of these outputs may be the same good; that is, firms  $j$  and  $j'$  may produce the same commodity.

these goods.<sup>10</sup> Output is sold in world markets. Factors, on the other hand, are immobile and must be used for production within the country.

The central question for our analysis concerns the equilibrium in the factor markets; that is, we wish to determine the equilibrium factor prices  $w = (w_1, \dots, w_L)$  and the allocation of the economy's factor endowments among the  $J$  firms.<sup>11</sup>

Given output prices  $p = (p_1, \dots, p_J)$  and input prices  $w = (w_1, \dots, w_L)$ , a profit-maximizing production plan for firm  $j$  solves

$$\begin{aligned} \text{Max} \quad & p_j f_j(z_j) - w \cdot z_j \\ \text{s.t.} \quad & z_j \geq 0 \end{aligned}$$

We denote firm  $j$ 's set of optimal input demands given prices  $(p, w)$  by  $z(p, w) \in \mathbb{R}_+^L$ . Because consumers have no direct use for their factor endowments, the total factor supply will be  $(\bar{z}_1, \dots, \bar{z}_L)$  as long as the input prices  $w_\ell$  are strictly positive (the only case that will concern us here). An equilibrium for the factor markets of this economy given the fixed output prices  $p$  therefore consists of an input price vector  $w^* = (w_1^*, \dots, w_L^*) \gg 0$  and a factor allocation

$$(z_1^*, \dots, z_J^*) = ((z_{11}^*, \dots, z_{L1}^*), \dots, (z_{1J}^*, \dots, z_{LJ}^*)),$$

such that firms receive their desired factor demands under prices  $(p, w^*)$  and all the factor markets clear, that is, such that

$$z_j^* \in z_j(p, w) \quad \text{for all } j = 1, \dots, J$$

and

$$\sum_j z_{\ell j}^* = \bar{z}_\ell \quad \text{for all } \ell = 1, \dots, L.$$

Because of the concavity of firms' production functions, first-order conditions are both necessary and sufficient for the characterization of optimal factor demands. Therefore, the  $L(J+1)$  variables formed by the factor allocation  $(z_1^*, \dots, z_J^*) \in \mathbb{R}_+^{LJ}$  and the factor prices  $w^* = (w_1^*, \dots, w_L^*)$  constitute an equilibrium if and only if they satisfy the following  $L(J+1)$  equations (we assume an interior solution here):

$$p_j \frac{\partial f_j(z_j^*)}{\partial z_{\ell j}} = w_\ell^* \quad \text{for } j = 1, \dots, J \text{ and } \ell = 1, \dots, L \quad (15.D.1)$$

and

$$\sum_j z_{\ell j}^* = \bar{z}_\ell \quad \text{for } \ell = 1, \dots, L. \quad (15.D.2)$$

The equilibrium output levels are then  $q_j^* = f_j(z_j^*)$  for every  $j$ .

Equilibrium conditions for *outputs* and factor prices can alternatively be stated using the firms' cost functions  $c_j(w, q_j)$  for  $j = 1, \dots, J$ . Output levels  $(q_1^*, \dots, q_J^*) \gg 0$  and factor prices  $w^* \gg 0$  constitute an equilibrium if and only if the following

10. See Exercise 15.D.4 for an endogenous determination (up to a scalar multiple) of the prices  $p = (p_1, \dots, p_J)$ .

11. Note that once the factor prices and allocations are determined, each consumer's demands can be readily determined from his demand function given the exogenous prices  $(p_1, \dots, p_J)$  and the wealth derived from factor input sales and profit distributions. Recall that the current model is completed by assuming that this demand is met in the world markets.

conditions hold:

$$p_j = \frac{\partial c_j(w^*, q_j^*)}{\partial q_j} \quad \text{for } j = 1, \dots, J, \quad (15.D.3)$$

$$\sum_j \frac{\partial c_j(w^*, q_j^*)}{\partial w_\ell} = \bar{z}_\ell \quad \text{for } \ell = 1, \dots, L. \quad (15.D.4)$$

Conditions (15.D.3) and (15.D.4) constitute a system of  $L + J$  equations in the  $L + J$  endogenous variables  $(w_1, \dots, w_L)$  and  $(q_1, \dots, q_J)$ . Condition (15.D.3) states that each firm must be at a profit-maximizing output level given prices  $p$  and  $w^*$ . If so, firm  $j$ 's optimal demand for the  $\ell$ th input is  $z_{\ell j}^* = \partial c_j(w, q_j^*) / \partial w_\ell$  (this is Shepard's lemma; see Proposition 5.C.2). Condition (15.D.4) is therefore the factor market-clearing condition.

Before examining the determinants of the equilibrium factor allocation in greater detail, we note that the equilibrium factor allocation  $(z_1^*, \dots, z_J^*)$  in this model is exactly the factor allocation that would be chosen by a revenue-maximizing planner, thus providing us with yet another expression of the welfare-maximizing property of competitive allocations (the first welfare theorem).<sup>12</sup> To see this, consider the problem faced by a planning authority who is charged with coordinating factor allocations for the economy in order to maximize the gross revenues from the economy's production activities:

$$\begin{aligned} \text{Max}_{(z_1, \dots, z_J) \geq 0} \quad & \sum_j p_j f_j(z_j) \\ \text{s.t.} \quad & \sum_j z_j = \bar{z}. \end{aligned} \quad (15.D.5)$$

How does the equilibrium factor allocation  $(z_1^*, \dots, z_J^*)$  compare with what this planner does? Recall from Section 5.E that whenever we have a collection of  $J$  price-taking firms, their profit-maximizing behavior is compatible with the behavior we would observe if the firms were to maximize their profits jointly taking the prices of outputs and factors as given. That is, the factor demands  $(z_1^*, \dots, z_J^*)$  solve

$$\text{Max}_{(z_1, \dots, z_J) \geq 0} \quad \sum_j (p_j f_j(z_j) - w^* \cdot z_j). \quad (15.D.6)$$

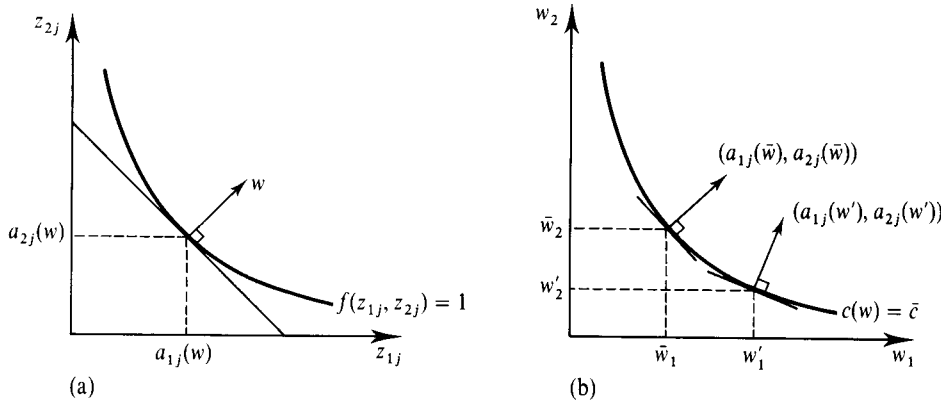
Since  $\sum_j z_j^* = \bar{z}$  (by the equilibrium property of market clearing), the factor demands  $(z_1^*, \dots, z_J^*)$  must also solve problem (15.D.6) subject to the further constraint that  $\sum_j z_j = \bar{z}$ . But this implies that the factor demands  $(z_1^*, \dots, z_J^*)$  in fact solve problem (15.D.5): if we must have  $\sum_j z_j = \bar{z}$ , then the total cost  $w^* \cdot (\sum_j z_j)$  is given, and so the joint profit-maximizing problem (15.D.6) reduces to the revenue-maximizing problem (15.D.5).

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One benefit of the property just established is that it can be used to obtain the equilibrium factor allocation without a previous explicit computation of the equilibrium factor prices; we simply need to solve problem (15.D.5) directly. It also provides a useful way of viewing the equilibrium factor prices. To see this, consider again the joint profit-maximization problem (15.D.6). We can approach this problem in an equivalent manner by first deriving an aggregate

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12. Note that maximization of economy-wide revenue from production would be the goal of any planner who wanted to maximize consumer welfare: it allows for the maximal purchases of consumption goods, at the fixed world prices.



**Figure 15.D.1**  
(a) A unit isoquant.  
(b) The unit cost function.

production function for dollars:

$$f(z) = \underset{(z_1, \dots, z_J) \geq 0}{\text{Max}} \quad p_1 f_1(z_1) + \dots + p_J f_J(z_J) \\ \text{s.t. } \sum_j z_j = z.$$

The aggregate factor demands must then solve  $\text{Max}_{z \geq 0} (f(z) - w \cdot z)$ . For every  $\ell$ , the first-order condition for this problem is  $w_\ell = \partial f(z)/\partial z_\ell$ . Moreover, at an equilibrium, the aggregate usage of factor  $\ell$  must be exactly  $\bar{z}_\ell$ . Hence, the equilibrium factor price of factor  $\ell$  must be  $w_\ell = \partial f(\bar{z})/\partial z_\ell$ ; that is, *the price of factor  $\ell$  must be exactly equal to its aggregate marginal productivity (in terms of revenue)*. Since  $f(\cdot)$  is concave, this observation by itself generates some interesting comparative statics. For example, a change in the endowment of a single input must change the equilibrium price of the input in the opposite direction.

Let us now be more specific and take  $J = L = 2$ , so that the economy under study produces two outputs from two primary factors. We also assume that the production functions  $f_1(z_{11}, z_{21})$ ,  $f_2(z_{12}, z_{22})$  are homogeneous of degree one (so the technologies exhibit constant returns to scale; see Section 5.B). This model is known as the  $2 \times 2$  production model. In applications, factor 1 is often thought of as labor and factor 2 as capital.

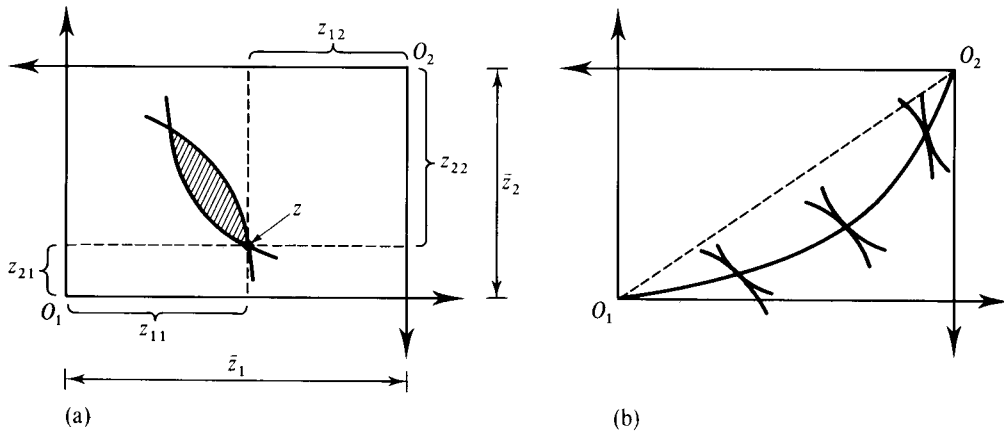
For every vector of factor prices  $w = (w_1, w_2)$ , we denote by  $c_j(w)$  the minimum cost of producing one unit of good  $j$  and by  $a_j(w) = (a_{1j}(w), a_{2j}(w))$  the input combination (assumed unique) at which this minimum cost is reached. Recall again from Proposition 5.C.2 that  $\nabla c_j(w) = (a_{1j}(w), a_{2j}(w))$ .

Figure 15.D.1(a) depicts the unit isoquant of firm  $j$ ,

$$\{(z_{1j}, z_{2j}) \in \mathbb{R}_+^2 : f_j(z_{1j}, z_{2j}) = 1\},$$

along with the cost-minimizing input combination  $(a_{1j}(w), a_{2j}(w))$ . In Figure 15.D.1(b), we draw a level curve of the unit cost function,  $\{(w_1, w_2) : c_j(w_1, w_2) = \bar{c}\}$ . This curve is downward sloping because as  $w_1$  increases,  $w_2$  must fall in order to keep the minimized costs of producing one unit of good  $j$  unchanged. Moreover, the set  $\{(w_1, w_2) : c_j(w_1, w_2) \geq \bar{c}\}$  is convex because of the concavity of the cost function  $c_j(w)$  in  $w$ . Note that the vector  $\nabla c_j(\bar{w})$ , which is normal to the level curve at  $\bar{w} = (\bar{w}_1, \bar{w}_2)$ , is exactly  $(a_{1j}(\bar{w}), a_{2j}(\bar{w}))$ . As we move along the curve toward higher  $w_1$  and lower  $w_2$ , the ratio  $a_{1j}(w)/a_{2j}(w)$  falls.

Consider, first, the efficient factor allocations for this model. In Figure 15.D.2, we



**Figure 15.D.2** (a) An inefficient factor allocation. (b) The Pareto set of factor allocations.

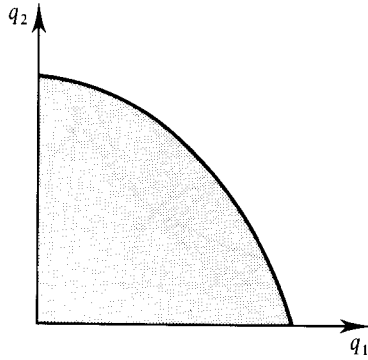
represent the possible allocations of the factor endowments between the two firms in an Edgeworth box of size  $\bar{z}_1$  by  $\bar{z}_2$ . The factors used by firm 1 are measured from the southwest corner; those used by firm 2 are measured from the northeast corner. We also represent the isoquants of the two firms in this Edgeworth box. Figure 15.D.2(a) depicts an inefficient allocation  $z$  of the inputs between the two firms: Any allocation in the interior of the hatched region generates more output of *both* goods than does  $z$ . Figure 15.D.2(b), on the other hand, depicts the Pareto set of factor allocations, that is, the set of factor allocations at which it is not possible, with the given total factor endowments, to produce more of one good without producing less of the other.

The Pareto set (endpoints excluded) must lie all above or all below or be coincident with the diagonal of the Edgeworth box. If it ever cuts the diagonal then because of constant returns, the isoquants of the two firms must in fact be tangent all along the diagonal, and so the diagonal must be the Pareto set (see also Exercise 15.B.7). Moreover, you should convince yourself of the correctness of the following claims.

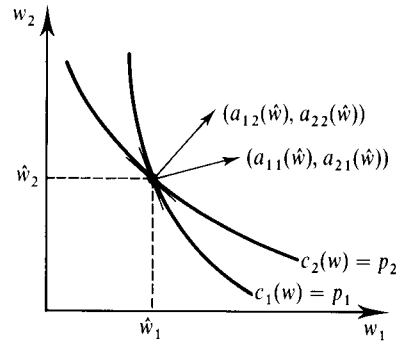
**Exercise 15.D.1:** Suppose that the Pareto set of the  $2 \times 2$  production model does not coincide with the diagonal of the Edgeworth box.

- Show that in this case, the factor intensity (the ratio of a firm's use of factor 1 relative to factor 2) of one of the firms exceeds that of the other at every point along the Pareto set.
- Show that in this case, any ray from the origin of either of the firms can intersect the Pareto set at most once. Conclude that the factor intensities of the two firms and the supporting relative factor prices change monotonically as we move along the Pareto set from one origin to the other.

In Figure 15.D.3, we depict the set of nonnegative output pairs  $(q_1, q_2)$  that can be produced using the economy's available factor inputs. This set is known as the *production possibility set*. Output pairs on the frontier of this set arise from factor allocations lying in the Pareto set of Figure 15.D.2(b). (Exercise 15.D.2 asks you to prove that the production possibility set is convex, as shown in Figure 15.D.3.)



**Figure 15.D.3 (left)**  
The production possibility set.



**Figure 15.D.4 (right)**  
The equilibrium factor prices and factor intensities in an interior equilibrium.

With the purpose of examining more closely the determinants of the equilibrium factor allocation  $(z_1^*, z_2^*)$  and the corresponding equilibrium factor prices  $w^* = (w_1^*, w_2^*)$ , we now assume that the *factor intensities* of the two firms bear a systematic relation to one another. In particular, we assume that in the production of good 1, there is, relative to good 2, a greater need for the first factor. In Definition 15.D.1 we make precise the meaning of “greater need”.

**Definition 15.D.1:** The production of good 1 is *relatively more intensive in factor 1* than is the production of good 2 if

$$\frac{a_{11}(w)}{a_{21}(w)} > \frac{a_{12}(w)}{a_{22}(w)}$$

at *all* factor prices  $w = (w_1, w_2)$ .

To determine the equilibrium factor prices, suppose that we have an *interior* equilibrium in which the production levels of the two goods are strictly positive (otherwise, we say that the equilibrium is *specialized*). Given our constant returns assumption, a necessary condition for  $(w_1^*, w_2^*)$  to be the factor prices in an interior equilibrium is that it satisfies the system of equations

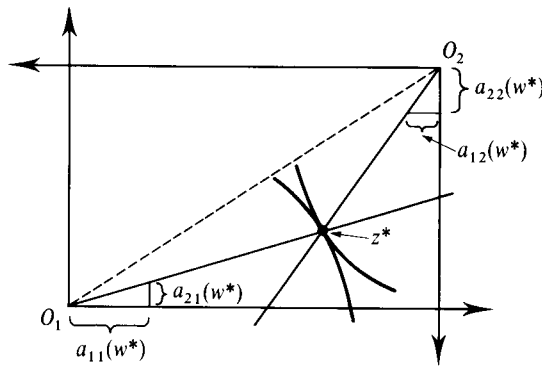
$$c_1(w_1, w_2) = p_1 \quad \text{and} \quad c_2(w_1, w_2) = p_2. \quad (15.D.7)$$

That is, at an interior equilibrium, prices must be equal to unit cost. This gives us two equations for the two unknown factor prices  $w_1$  and  $w_2$ .<sup>13</sup>

Figure 15.D.4 depicts the two unit cost functions in (15.D.7). By expression (15.D.7), a necessary condition for  $(\hat{w}_1, \hat{w}_2)$  to be the factor prices of an interior equilibrium is that these curves cross at  $(\hat{w}_1, \hat{w}_2)$ . Moreover, the factor intensity assumption implies that whenever the two curves cross, the curve for firm 2 must be flatter (less negatively sloped) than that for firm 1 [recall that  $\nabla c_j(w) = (a_{1j}(w), a_{2j}(w))$ ]. From this, it follows that the two curves can cross at most once.<sup>14</sup> Hence, under the

13. Expression (15.D.7) is the constant returns version of (15.D.3). Note that the effect of the constant returns to scale assumption is to make (15.D.3) independent of the output levels  $(q_1, \dots, q_J)$  (for interior equilibria).

14. If they crossed several times, then the curve for firm 2 must cross the curve for firm 1 at least once from above. At this crossing point, the curve for firm 2 would be steeper than the curve for firm 1, contradicting the factor intensity condition.



**Figure 15.D.5**  
The equilibrium factor allocation.

factor intensity condition, there is at most a single pair of factor prices that can arise as the equilibrium factor prices of an interior equilibrium.<sup>15</sup>

Once the equilibrium factor prices  $w^*$  are known, the equilibrium output levels can be found graphically by determining the unique point  $(z_1^*, z_2^*)$  in the Edgeworth box of factor allocations at which both firms have the factor intensities associated with factor prices  $w^*$ , that is,

$$\frac{z_{11}^*}{z_{21}^*} = \frac{a_{11}(w^*)}{a_{21}(w^*)} \quad \text{and} \quad \frac{z_{12}^*}{z_{22}^*} = \frac{a_{12}(w^*)}{a_{22}(w^*)}.$$

The construction is depicted in Figure 15.D.5.

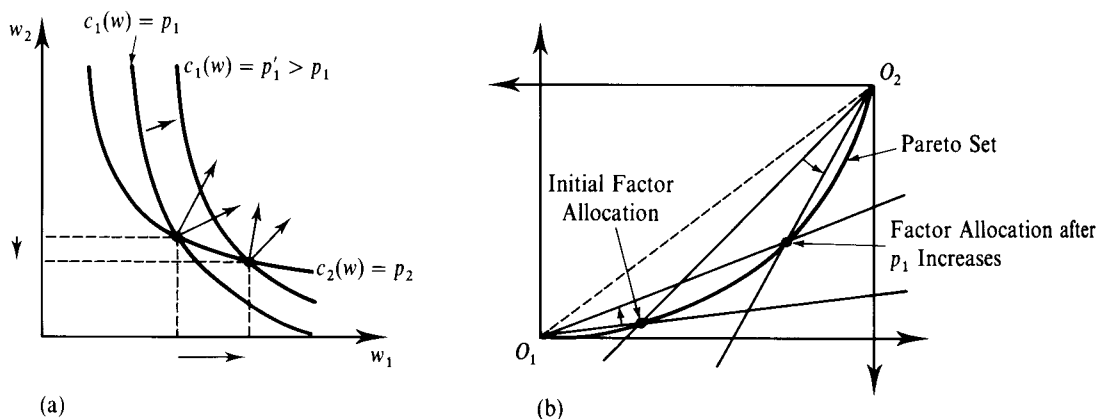
An important consequence of this discussion is that in the  $2 \times 2$  production model, if the factor intensity condition holds, then as long as the economy does not specialize in the production of a single good [and therefore (15.D.7) holds], the equilibrium factor prices depend *only on the technologies of the two firms and on the output prices  $p$* . Thus, the levels of the endowments matter only to the extent that they determine whether the economy specializes. This result is known in the international trade literature as the *factor price equalization theorem*. The theorem provides conditions (which include the presence of tradable consumption goods, identical production technologies in each country, and price-taking behavior) under which the prices of nontradable factors are equalized across nonspecialized countries.

We now present two comparative statics exercises. We first ask: How does a change in the price of one of the outputs, say  $p_1$ , affect the equilibrium factor prices and factor allocations? Figure 15.D.6(a), which depicts the induced change in Figure 15.D.4, identifies the change in factor prices. The increase in  $p_1$  shifts firm 1's curve

15. Note, however, that although  $(\hat{w}_1, \hat{w}_2)$  may solve (15.D.7), this is not sufficient to ensure that  $(\hat{w}_1, \hat{w}_2)$  are equilibrium factor prices. In particular, even though  $(\hat{w}_1, \hat{w}_2)$  solve (15.D.7), no interior equilibrium may exist. In Exercise 15.D.6, you are asked to show that under the factor intensity condition, the equilibrium will involve positive production of the two goods if and only if

$$\frac{a_{11}(\hat{w})}{a_{21}(\hat{w})} > \frac{\bar{z}_1}{\bar{z}_2} > \frac{a_{12}(\hat{w})}{a_{22}(\hat{w})},$$

where  $\hat{w} = (\hat{w}_1, \hat{w}_2)$  is the unique solution to (15.D.7). In words, the factor intensity of the overall economy must be intermediate between the factor intensities of the two firms computed at the sole vector of factor prices at which diversification can conceivably occur.



**Figure 15.D.6** The Stolper-Samuelson theorem. (a) The change in equilibrium factor prices. (b) The change in the equilibrium factor allocation.

[the set  $\{(w_1, w_2): c_1(w_1, w_2) = p_1\}$ ] outward toward higher factor price levels; the point of intersection of the two curves moves out along firm 2's curve to a higher level of  $w_1$  and a lower level of  $w_2$ .

Formally, this gives us the result presented in Proposition 15.D.1.

**Proposition 15.D.1: (Stolper-Samuelson Theorem)** In the  $2 \times 2$  production model with the factor intensity assumption, if  $p_j$  increases, then the equilibrium price of the factor more intensively used in the production of good  $j$  increases, while the price of the other factor decreases (assuming interior equilibria both before and after the price change).<sup>16</sup>

**Proof:** For illustrative purposes, we provide a formal proof to go along with the graphical analysis of Figure 15.D.6 presented above. Note that it suffices to prove the result for an infinitesimal change  $dp = (1, 0)$ .

Differentiating conditions (15.D.7), we have

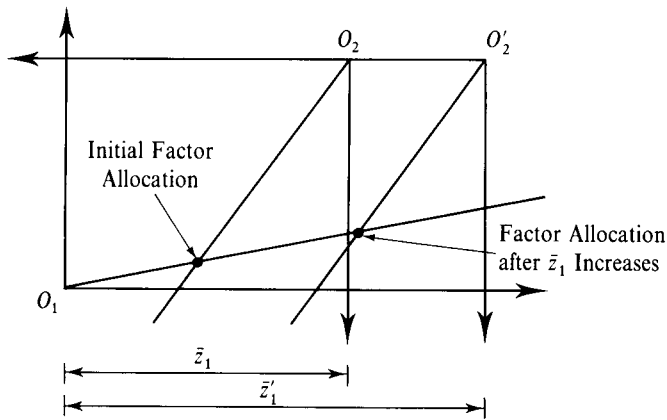
$$dp_1 = \nabla c_1(w^*) \cdot dw = a_{11}(w^*) dw_1 + a_{21}(w^*) dw_2,$$

$$dp_2 = \nabla c_2(w^*) \cdot dw = a_{12}(w^*) dw_1 + a_{22}(w^*) dw_2,$$

or in matrix notation,

$$dp = \begin{bmatrix} a_{11}(w^*) & a_{21}(w^*) \\ a_{12}(w^*) & a_{22}(w^*) \end{bmatrix} dw.$$

16. See Exercise 15.D.3 for a strengthening of this conclusion. We also note that, strictly speaking, the factor intensity condition is not required for this result. The reason is that, as we saw in Exercise 15.D.1, the firm that uses one factor, say factor 1, more intensively is the same for any point in the Pareto set of factor allocations. Suppose, for example, that we are as in Figure 15.D.2(b), where firm 1 uses factor 1 more intensively. Then, when  $p_1$  rises, we can see from Figure 15.D.3, and the overall revenue-maximizing property of equilibrium discussed earlier in this section, that the output of good 1 increases and that of good 2 decreases. This implies that we move along the Pareto set in Figure 15.D.2(b) toward firm 2's origin. Therefore, recalling Exercise 15.D.1, both firms' intensity of use of factor 1 decreases. Hence, the equilibrium factor price ratio  $w_1^*/w_2^*$  must increase. Finally, since firm 2 is still breaking even and its output price has not changed, this implies that  $w_1^*$  increases and  $w_2^*$  decreases.



**Figure 15.D.7**  
The Rybczynski theorem.

Denote this  $2 \times 2$  matrix by  $A$ . The factor intensity assumption implies that  $|A| = a_{11}(w^*)a_{22}(w^*) - a_{12}(w^*)a_{21}(w^*) > 0$ . Therefore  $A^{-1}$  exists and we can compute it to be

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} a_{22}(w^*) & -a_{21}(w^*) \\ -a_{12}(w^*) & a_{11}(w^*) \end{bmatrix}.$$

Hence, the entries of  $A^{-1}$  are positive at the diagonal and negative off the diagonal. Since  $dw = A^{-1} dp$ , this implies that for  $dp = (1, 0)$  we have  $dw_1 > 0$  and  $dw_2 < 0$ , as we wanted. ■

We have just seen that if  $p_1$  increases, then  $w_1^*/w_2^*$  increases. Therefore, both firms must move to a less intensive use of factor 1. Figure 15.D.6(b) depicts the resulting change in the equilibrium allocation of factors. As can be seen, the factor allocation moves to a new point in the Pareto set at which the output of good 1 has risen and that of good 2 has fallen.

For the second comparative statics exercise, suppose that the total availability of factor 1 increases from  $\bar{z}_1$  to  $\bar{z}'_1$ . What is the effect of this on equilibrium factor prices and output levels? Because neither the output prices nor the technologies have changed, the factor input prices remain unaltered (as long as the economy does not specialize). As a result, factor intensities also do not change. The new input allocation is then easily determined in the superimposed Edgeworth boxes of Figure 15.D.7; we merely find the new intersection of the two rays associated with the unaltered factor intensity levels.

Thus, examination of Figure 15.D.7 gives us the result presented in Proposition 15.D.2.

**Proposition 15.D.2: (Rybczynski Theorem)** In the  $2 \times 2$  production model with the factor intensity assumption, if the endowment of a factor increases, then the production of the good that uses this factor relatively more intensively increases and the production of the other good decreases (assuming interior equilibria both before and after the change of endowment).

For further discussion of the  $2 \times 2$  production model see, for example, Johnson (1971).

Consider the general case of an arbitrary number of factors  $L$  and outputs  $J$ . For given output prices, the zero-profit conditions [i.e., the general analog of expression (15.D.7)]

constitute a (nonlinear) system of  $J$  equations in  $L$  unknowns. If  $L > J$ , then there are too many unknowns and we cannot hope that the zero-profit conditions alone will determine the factor prices. The total factor endowments will play a role. If  $J > L$ , then there are too many equations and, for typical world prices, they cannot all be satisfied simultaneously. What this means is that the economy will specialize in the production of a number of goods equal to the number of factors  $L$ . The set of goods chosen may well depend on the endowments of factors. Beyond the  $2 \times 2$  situation (the analysis of which, as we have seen, is quite instructive), the case  $L = J$  seems too coincidental to be of interest. Nevertheless, we point out that in this case the zero-profit conditions are nonlinear and that in order to guarantee a unique solution (and versions of the Stolper–Samuelson and the Rybczynski theorems), we need a generalization of the factor intensity condition. These generalizations exist, but they cannot be interpreted economically in as simple a manner as can the factor intensity condition of the  $2 \times 2$  model.

## 15.E General Versus Partial Equilibrium Theory

There are some problems that are inherently general equilibrium problems. It would be hard to envision convincing analyses of economic growth, demographic change, international economic relations, or monetary policy that were restricted to only a subset of commodities and did not consider economy-wide feedback effects.

Partial equilibrium models of markets, or of systems of related markets, determine prices, profits, productions, and the other variables of interest adhering to the assumption that there are no feedback effects from these endogenous magnitudes to the underlying demand or cost curves that are specified in advance. Individuals' wealth is another variable that general equilibrium theory regards as endogenously determined but that is often treated as exogenous in partial equilibrium theory.

If general equilibrium analysis did not change any of the predictions or conclusions of partial equilibrium analysis, it would be of limited significance when applied to problems amenable to partial equilibrium treatment. It might be of comfort because we would then know that our partial equilibrium conclusions are valid, but it would not change our view of how markets work. However, things are not that simple. The choice of methodology may be far from innocuous. We now present an example [due to Bradford (1978)] in which a naive application of partial equilibrium analysis leads us seriously astray. See Sections 3.I and 10.G for some discussion of when partial equilibrium theory is (approximately) justified.

### *A Tax Incidence Example*

Consider an economy with a large number of towns,  $N$ . Each town has a single price-taking firm that produces a consumption good by means of the strictly concave production function  $f(z)$  (once again, we could reinterpret the model as having many identical firms in each town to make the price-taking hypothesis more palatable). This consumption good, which is identical across towns, is traded in a national market. The overall economy has  $M$  units of labor, inelastically supplied by workers who derive utility only from the output of the firms. Workers are free to move from town to town and do so to seek the highest wage. We normalize the price of the consumption good to be 1, and we denote the wage rate in town  $n$ 's labor market by  $w_n$ .

Given that workers can move freely in search of the highest wage, at an equilibrium the wage rates across towns must be equal; that is, we must have  $w_1 = \dots = w_N = \bar{w}$ . From the symmetry of the problem, it must be that each firm hires exactly  $M/N$  units of labor in an equilibrium. As a result, the equilibrium wage rate must be  $\bar{w} = f'(M/N)$ . The equilibrium profits of an individual firm are therefore  $f(M/N) - f'(M/N)(M/N)$ .

Now suppose that town 1 levies a tax on the labor used by the firm located there. We investigate the “incidence” of the tax on workers and firms (or, more properly, on the firms’ owners); that is, we examine the extent to which each group bears the burden of the tax. If the tax rate is  $t$  and the wage in town 1 is  $w_1$ , the labor demand of the firm in town 1 will be the amount  $z_1$  such that  $f'(z_1) = t + w_1$ . At this point, we may be tempted to argue that, since  $N$  is large, we can approximate and take the wage rates elsewhere,  $\bar{w}$ , to be unaffected by this change in town 1. Moreover, since labor moves freely, the supply correspondence of workers in town 1 should then be 0 at  $w_1 < \bar{w}$ ,  $\infty$  at  $w_1 > \bar{w}$ , and  $[0, \infty]$  at  $w_1 = \bar{w}$ . Thus, taking a partial equilibrium view, the equilibrium wage rate in the town 1 labor market remains equal to  $\bar{w}$ , and the labor employed in town 1 falls to the level  $z_1$  such that  $f'(z_1) = t + \bar{w}$  (hence, some labor will shift to the other towns). By adopting this sort of partial equilibrium view of the labor market of town 1, we are therefore led to conclude that the income of workers remains the same, as does the profit of every firm not located in town 1. Only the profit of the firm in town 1 decreases. The qualitative conclusion is that firms (actually, firms’ owners) “bear” all of the tax burden. Labor, because it is free to move and because the number of untaxed firms is large, “escapes.”

Alas, this conclusion constitutes an egregious mistake, and it will be overturned by a general equilibrium view of the same model.

We now look at the general equilibrium across the labor markets of all the towns. We know that the equilibrium wage rate must be such that  $w_1 = \dots = w_N$  and that all  $M$  units of labor are employed. Let  $w(t)$  be this common equilibrium wage when the tax rate in town 1 is  $t$ . By symmetry, the firms in towns 2,  $\dots$ ,  $N$  will each employ the same amount of labor,  $z(t)$ . Let  $z_1(t)$  be the equilibrium labor demand of the firm in town 1 when town 1’s tax rate is  $t$ . Then the equilibrium conditions are

$$(N - 1)z(t) + z_1(t) = M. \quad (15.E.1)$$

$$f'(z(t)) = w(t). \quad (15.E.2)$$

$$f'(z_1(t)) = w(t) + t. \quad (15.E.3)$$

Consider the impact on wages of the introduction of a small tax  $dt$ . Substituting from (15.E.1) for  $z_1(t)$  in (15.E.3), differentiating with respect to  $t$ , and evaluating at  $t = 0$  [at which point  $z_1(0) = z(0) = (M/N)$ ], we get

$$-f''(M/N)(N - 1)z'(0) = w'(0) + 1. \quad (15.E.4)$$

But from (15.E.2), we get

$$f''(M/N)z'(0) = w'(0). \quad (15.E.5)$$

Substituting from (15.E.5) into (15.E.4) yields

$$w'(0) = -\frac{1}{N}.$$

Therefore, once the general equilibrium effects are taken into account, we see that